Université: Institut Universitaire des Sciences (IUS)

TD N^0 3: Réseaux 2

Nom & Prénom: SAINT-JEAN Marc-Evenort

Professeur: Ismael SAINT AMOUR

Niveau: 3ème Année

Date: Mai 2026

Introduction

Ce travail consiste à configurer et sécuriser un réseau de campus universitaire avec Cisco Packet Tracer. L'objectif est de séparer les services en VLANs et de contrôler les accès entre eux pour garantir la sécurité du réseau.

Objectifs

Créer une topologie réseau (routeurs, switches, PCs).
Configurer des VLANs pour chaque service (admin, enseignants, étudiants, invités, serveurs).
Mettre en place le routage inter-VLAN.
Sécuriser le réseau avec des ACL, Port Security et SSH.
Tester la connectivité et la sécurité du réseau.

1. Topologie réseau

a) Réseau physique

Routeurs: R1, R2, R3

Switches: S1, S2, S3, S4, S5

PCs: PC1 à PC25

b) Répartition des PCs :

VLAN 10 : Administration (5 PCs)

VLAN 20 : Enseignants (5 PCs)

VLAN 30 : Étudiants (10 PCs)

VLAN 40 : WiFi invités (5 PCs)

VLAN 99 : Serveurs (DMZ)

Le switch principal (S1) est connecté au routeur R1, les autres switches sont interconnectés.

2. Schéma de la topologie

a) Création des VLANs

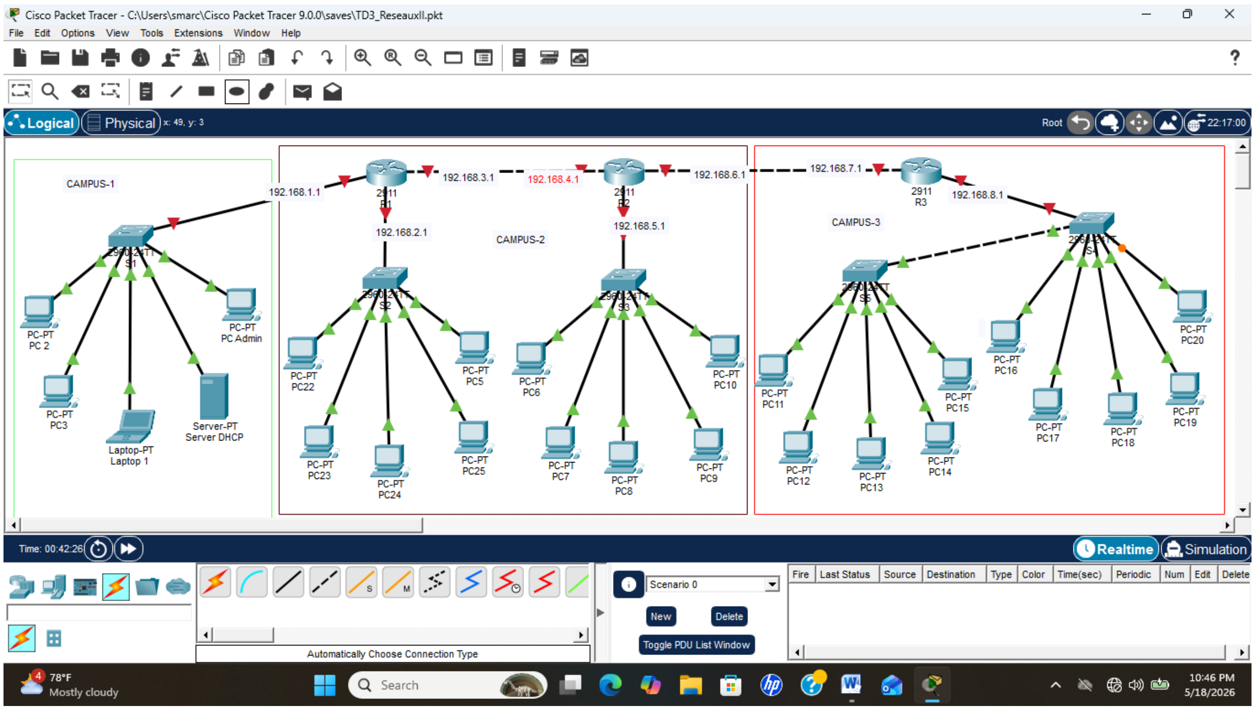


Figure 1: Cette image illustre la topologie du réseau demandée

3 Configuration réseau de base

1. VLANs sur switches

a)

Switch> enable
Switch# configure terminal

 # 1. Création des VLANs

Switch(config)# vlan 10
Switch(config-vlan)# name Admin
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Enseignants
Switch(config-vlan)# vlan 30

Switch(config-vlan)# name Etudiants
Switch(config-vlan)# vlan 40
Switch(config-vlan)# name Invites
Switch(config-vlan)# vlan 99
Switch(config-vlan)# name DMZ_Serveurs
Switch(config-vlan)# exit

 # 2. Assignation des ports d'accès (Ex: ports 1 à 5 pour l'Admin)
Switch(config)# interface range fa0/1 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

interface range fa0/6 - 10
switchport mode access
switchport access vlan 20
exit

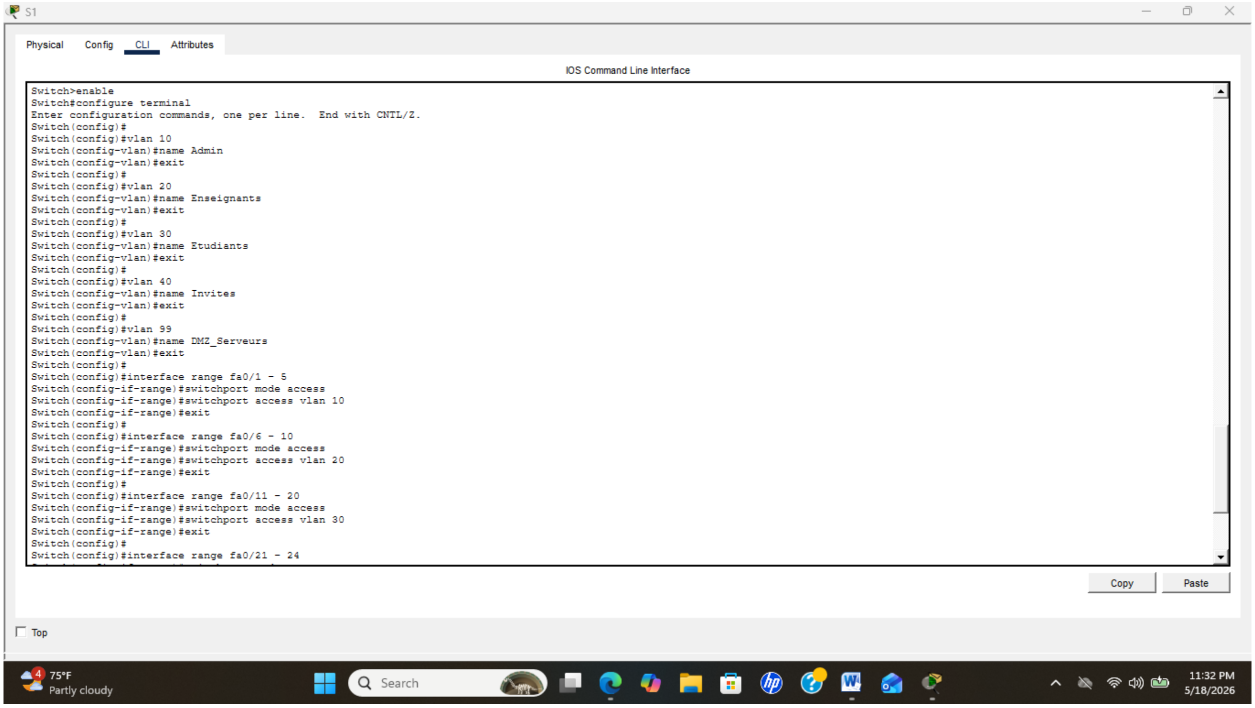
interface range fa0/11 - 20
switchport mode access
switchport access vlan 30
exit

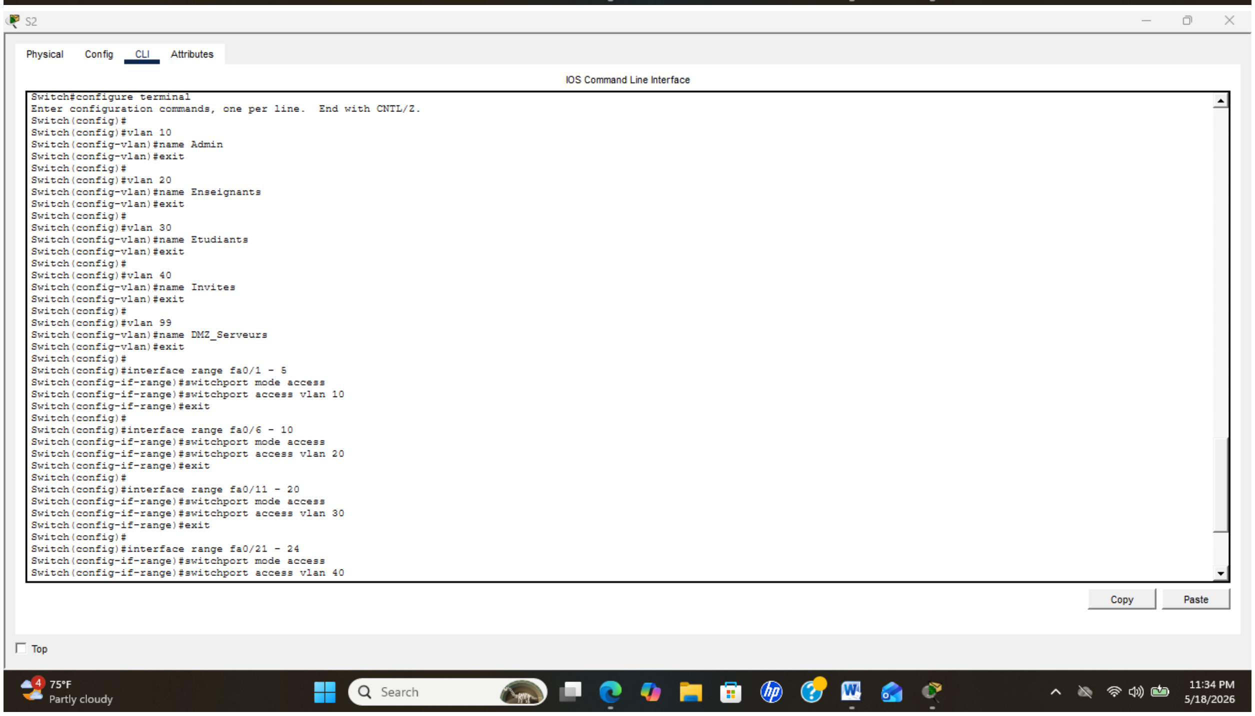
interface range fa0/21 - 24
switchport mode access
switchport access vlan 40
exit

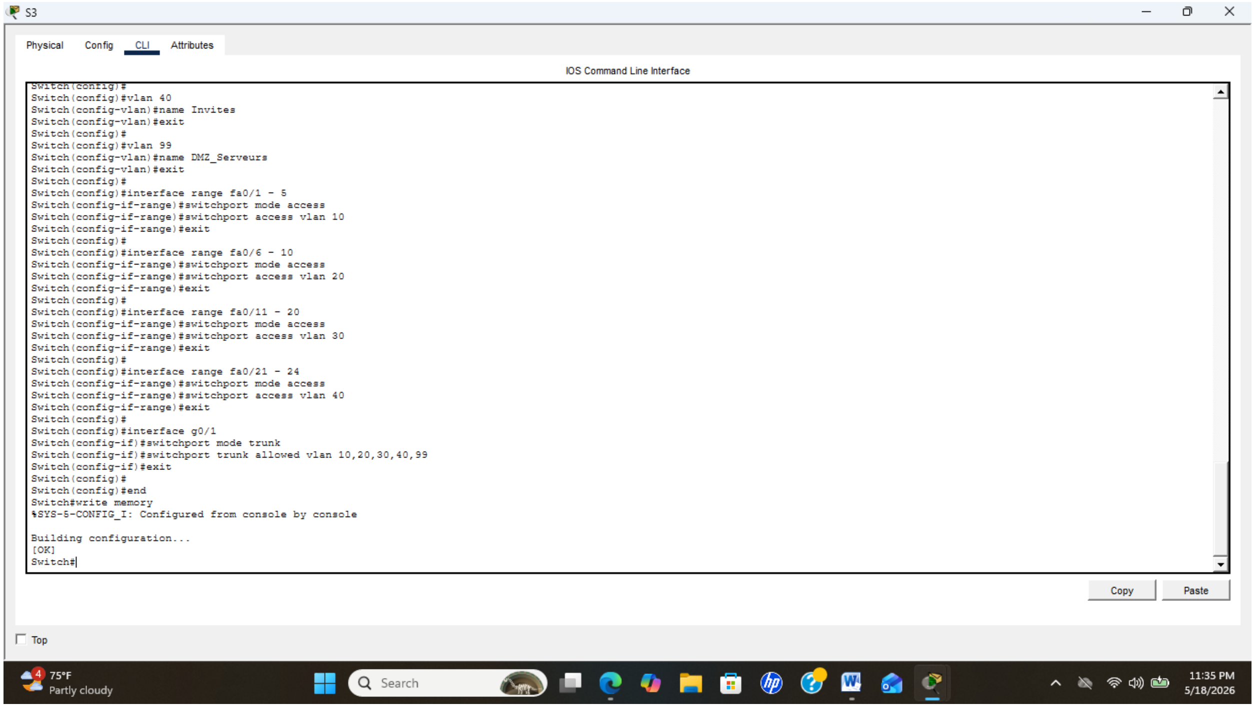
interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,99
exit

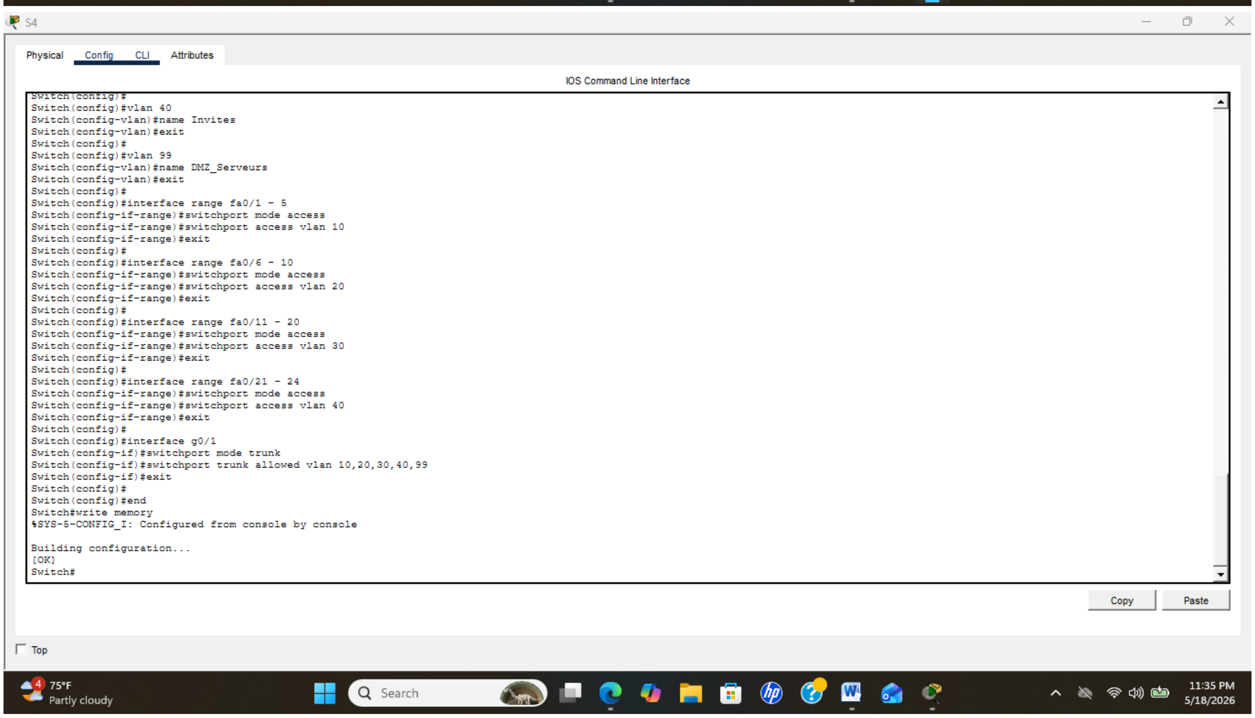
end

write memory






S3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#name Invites
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 99
Switch(config-vlan)#name DMZ_Serveurs
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface range fa0/1 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/6 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/11 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/21 - 24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Switch#
Copy
Paste
Top
75°F
Partly cloudy
Search
11:35 PM
5/18/2026


S4
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#name Invites
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 99
Switch(config-vlan)#name DMZ_Serveurs
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#interface range fa0/1 - 5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/6 - 10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/11 - 20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface range fa0/21 - 24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
Switch#
Copy
Paste
Top
75°F
Partly cloudy
Search
11:35 PM
5/18/2026

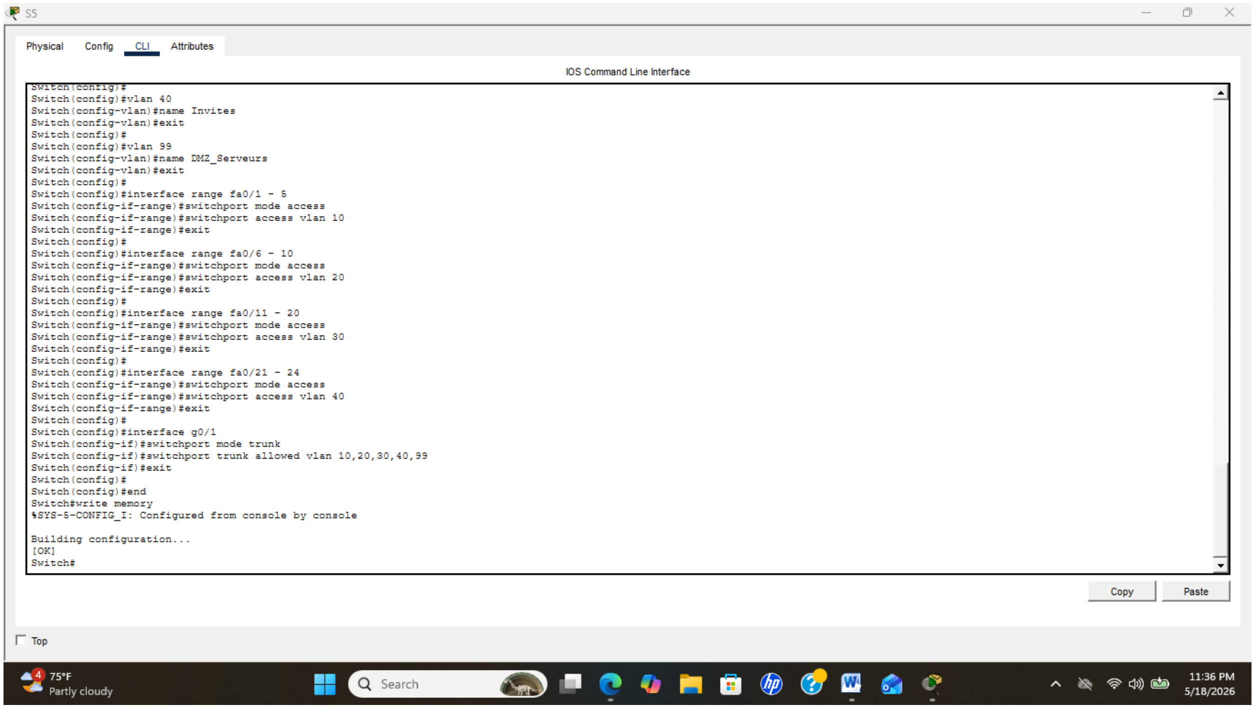


Figure 2 à 5: Ces images attestent la configuration des switches réalisée dans le cadre du TD (voir pages 5 à 7.

3.1 Attribution des adresses IP

Switch> enable

Switch# configure terminal

Switch(config)# interface vlan1

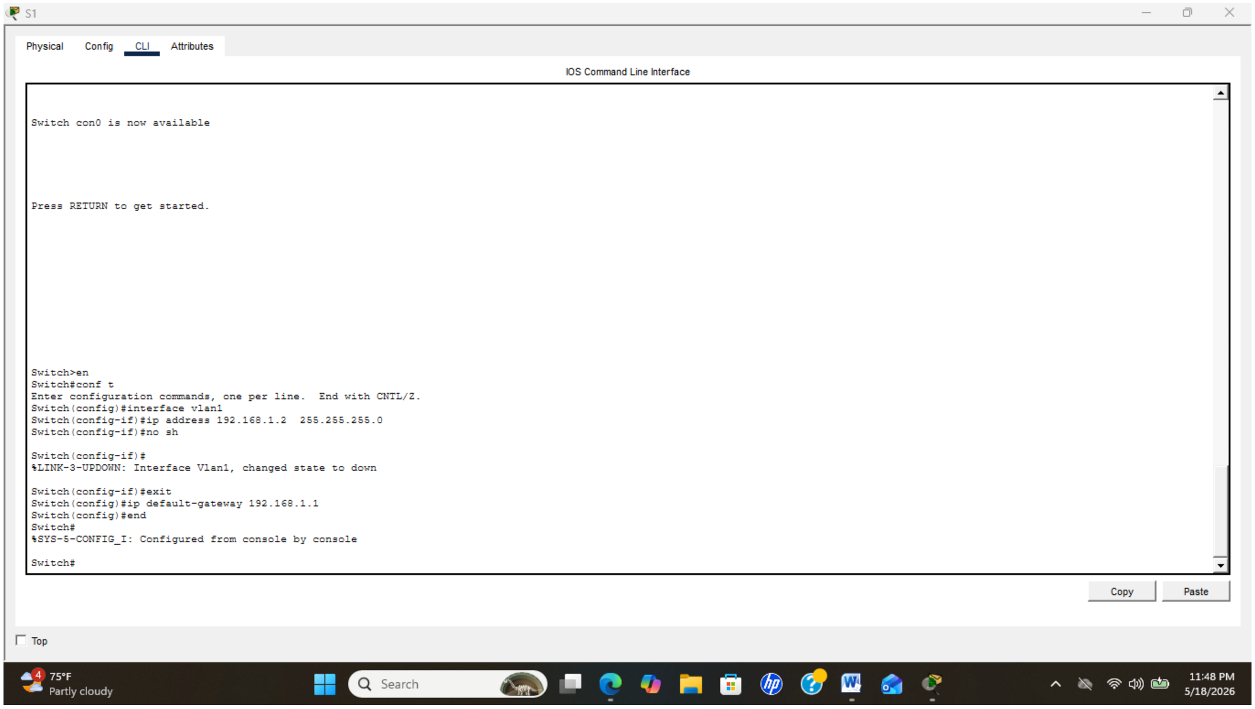
Switch(config)# ip address 192.168.1.2 255.255.255.0

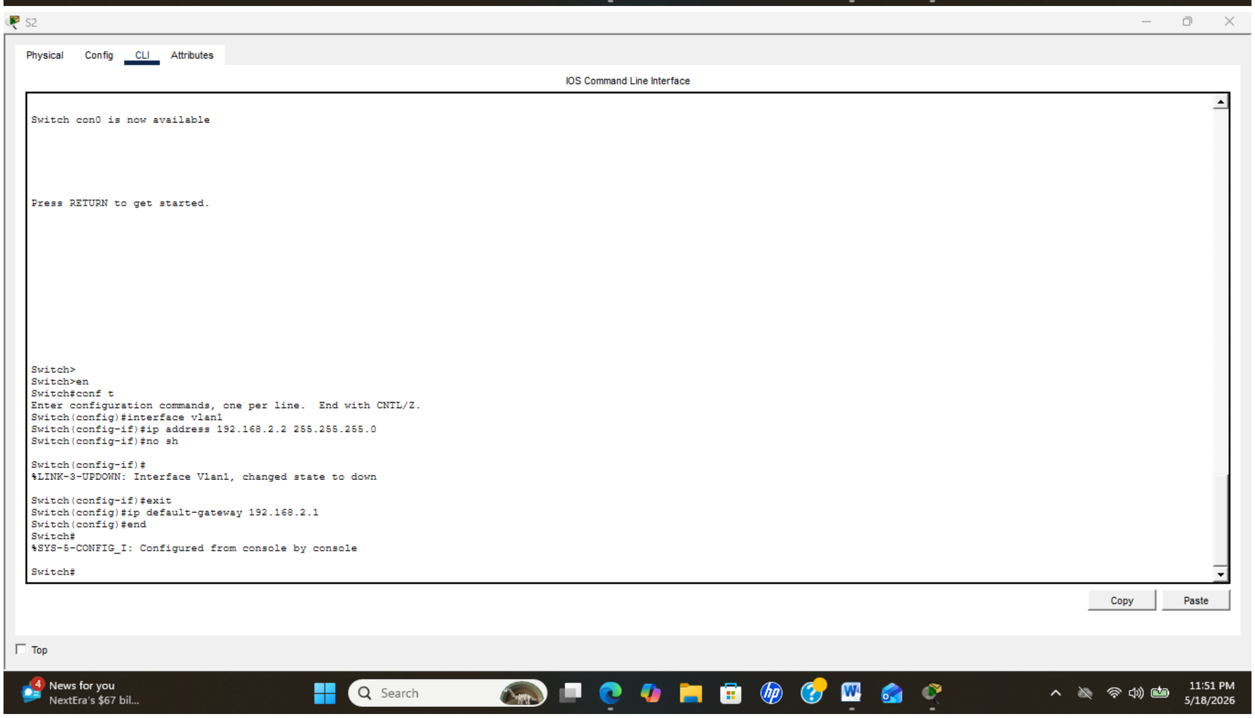
Switch(config)#no sh

Switch(config)#exit

Switch(config)#ip default-gateway 192.168.1.1

Switch(config)#end





Figures 6, 7 : Ces images montrent l'attribution des adresses IP aux différents équipements du réseau.

4 Configuration du Routage Inter-VLAN

Routeur 1

Router> enable
Router# configure terminal
Router(config)# hostname R1-Principal

R1-Principal(config)# interface gi 0/0
R1-Principal(config-if)# no shutdown
R1-Principal(config-if)# exit

 # --- SUB-INTERFACES INTER-VLAN ---

R1-Principal(config)# interface gi 0/0.10
R1-Principal(config-subif)# encapsulation dot1Q 10
R1-Principal(config-subif)# ip address 192.168.10.1 255.255.255.0
R1-Principal(config-subif)# exit

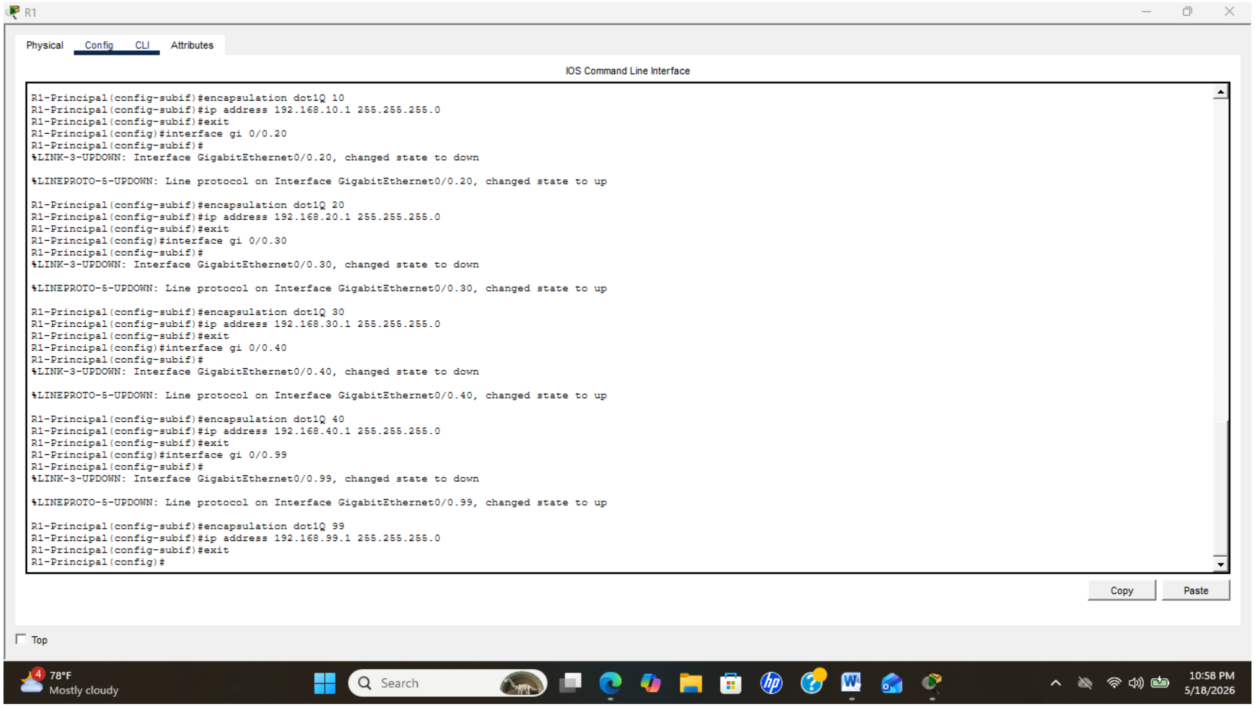
R1-Principal(config)# interface gi 0/0.20
R1-Principal(config-subif)# encapsulation dot1Q 20
R1-Principal(config-subif)# ip address 192.168.20.1 255.255.255.0
R1-Principal(config-subif)# exit

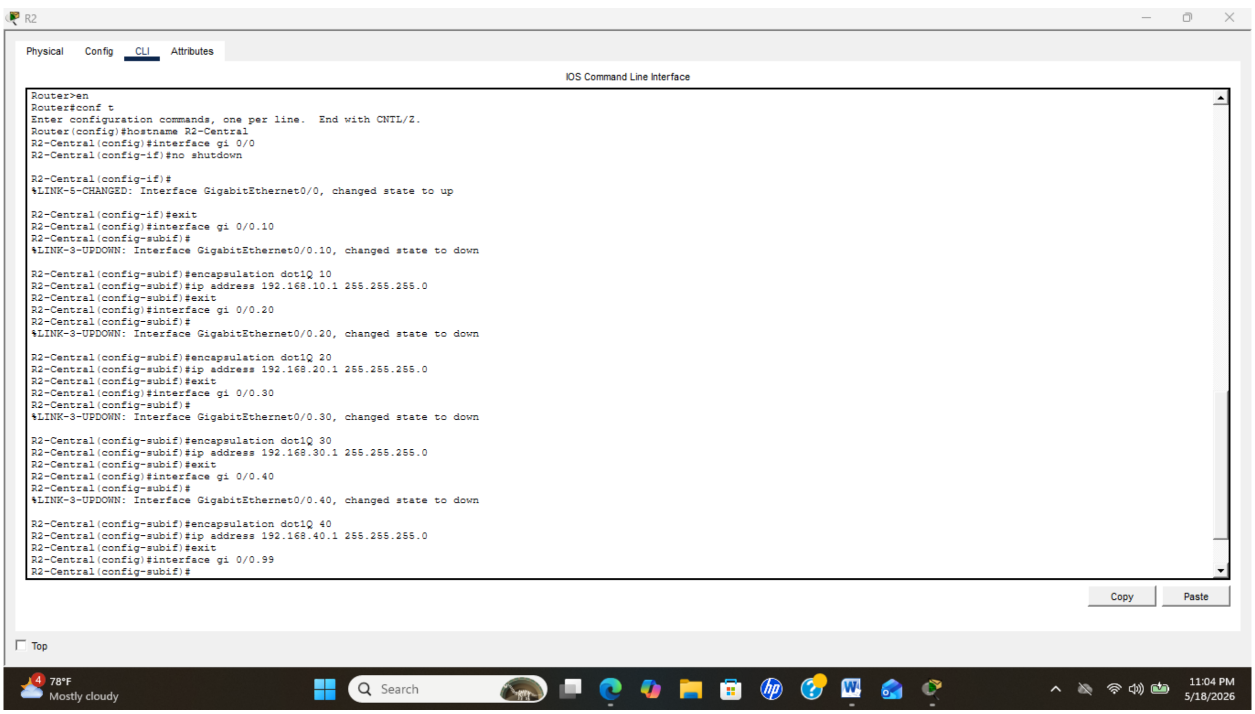
R1-Principal(config)# interface gi 0/0.30
R1-Principal(config-subif)# encapsulation dot1Q 30
R1-Principal(config-subif)# ip address 192.168.30.1 255.255.255.0
R1-Principal(config-subif)# exit

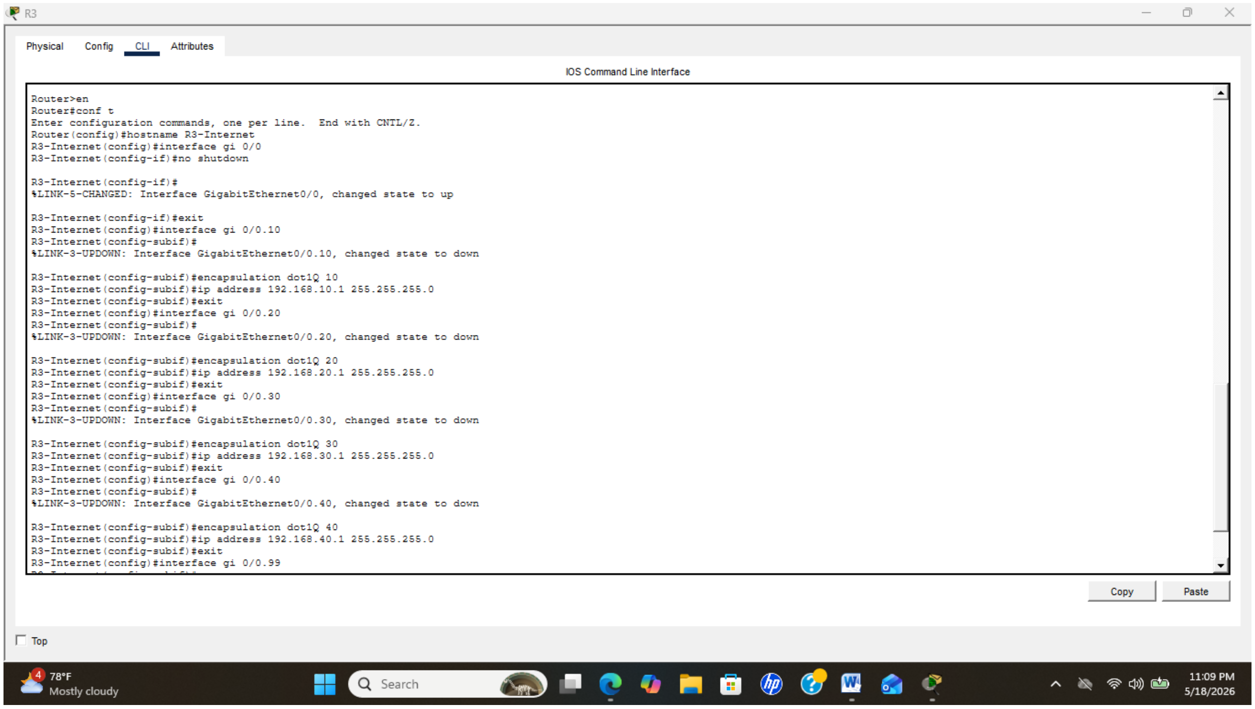
R1-Principal(config)# interface gi 0/0.40
R1-Principal(config-subif)# encapsulation dot1Q 40
R1-Principal(config-subif)# ip address 192.168.40.1 255.255.255.0
R1-Principal(config-subif)# exit

R1-Principal(config)# interface gi 0/0.99
R1-Principal(config-subif)# encapsulation dot1Q 99
R1-Principal(config-subif)# ip address 192.168.99.1 255.255.255.0

R1-Principal(config-subif)# exit

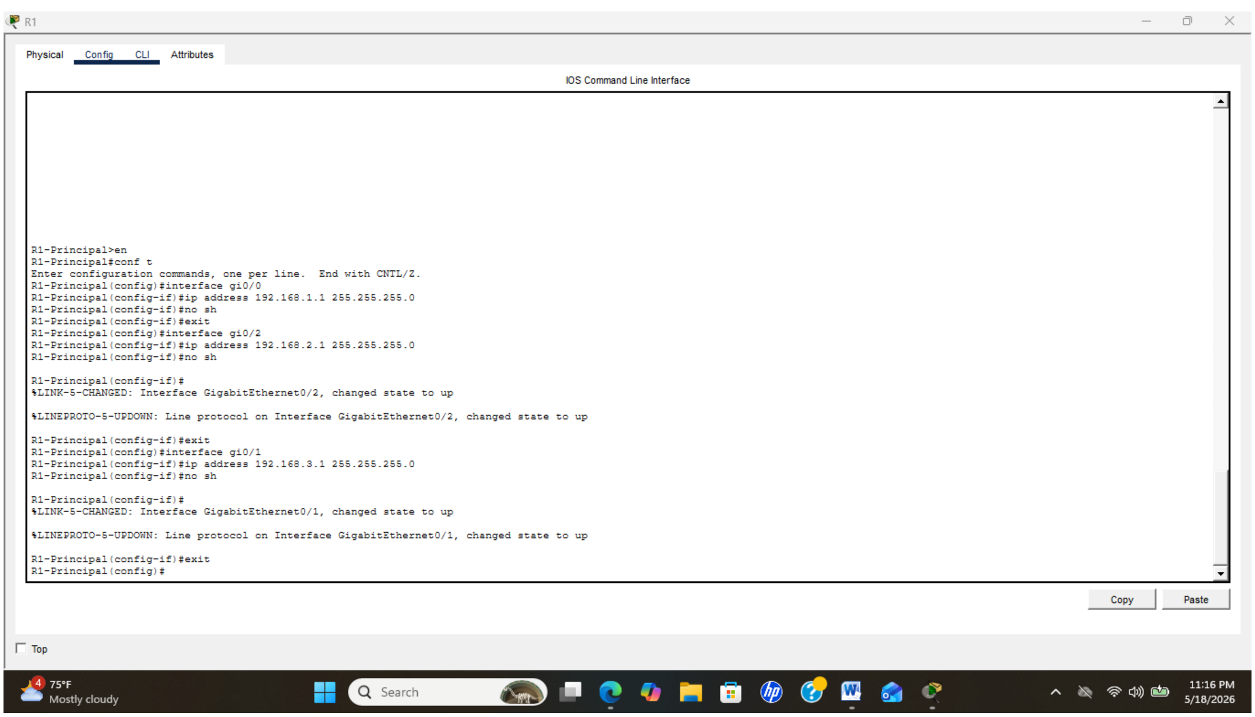


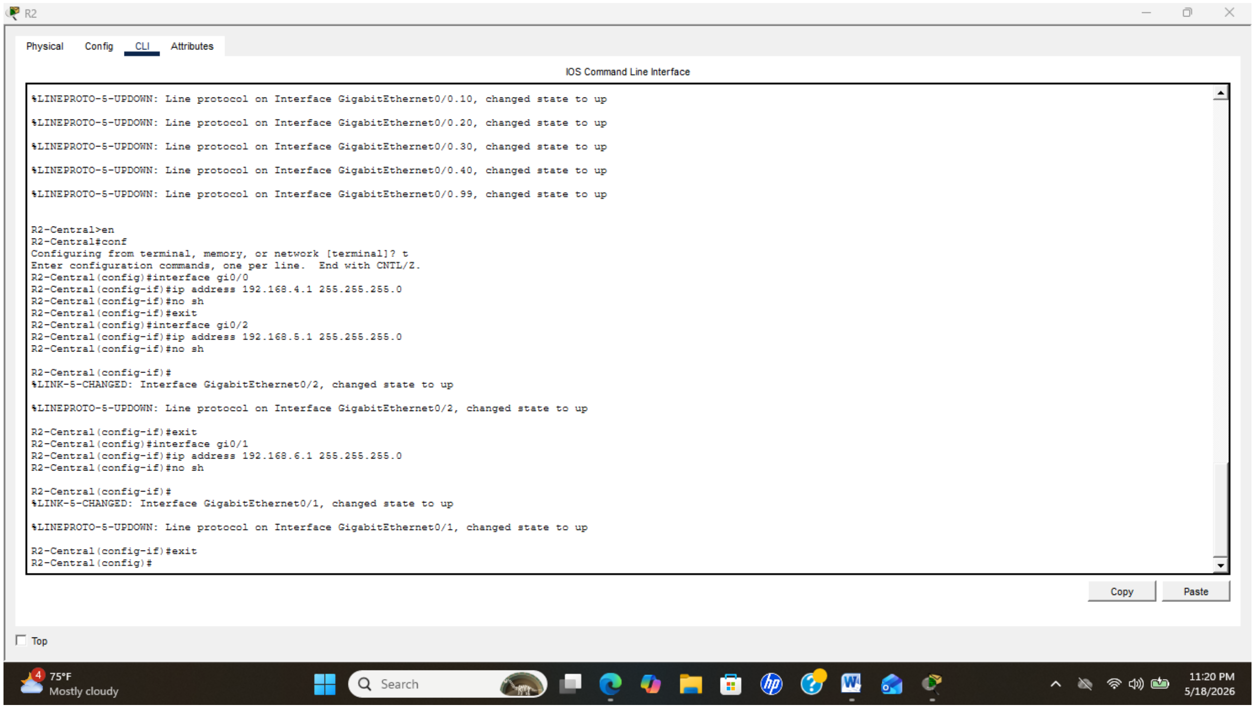


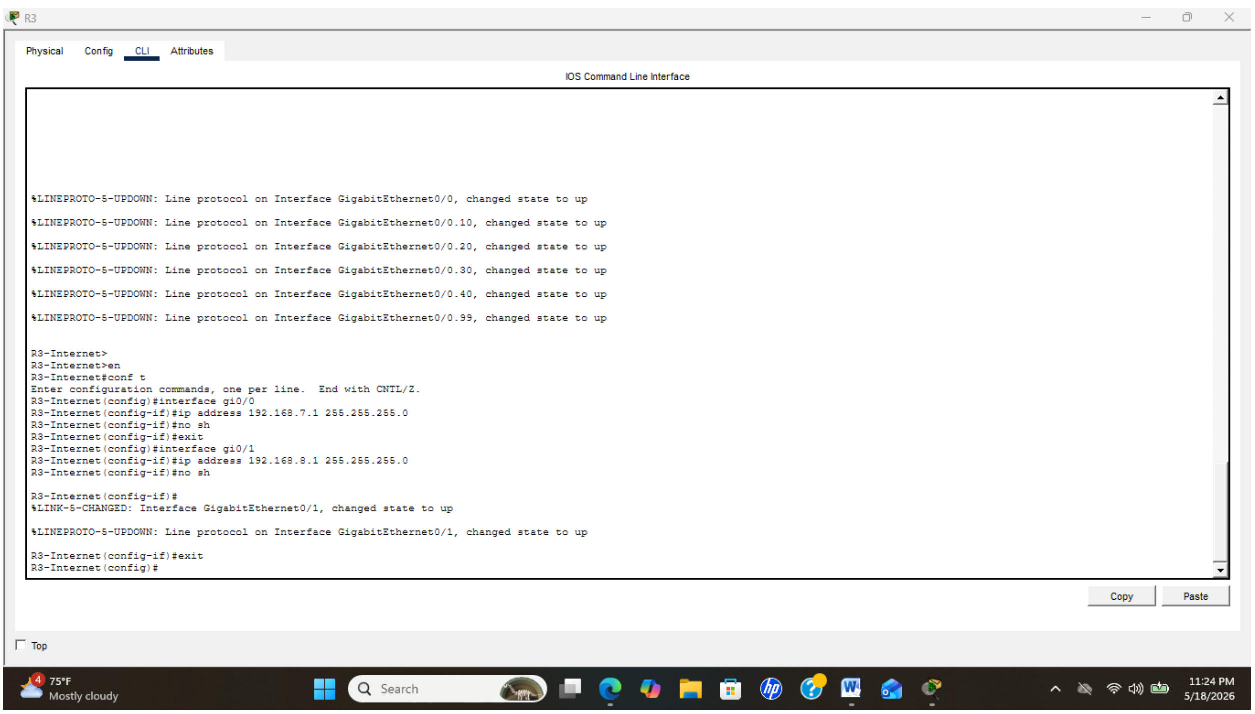


Figures 8 à 11 : Ces images montrent la configuration du routage dans le réseau.

4.1 Configuration des interfaces







Figures 11 à 14 : Ces images montrent la configuration des interfaces des routeurs (voir pages 11 à 12).

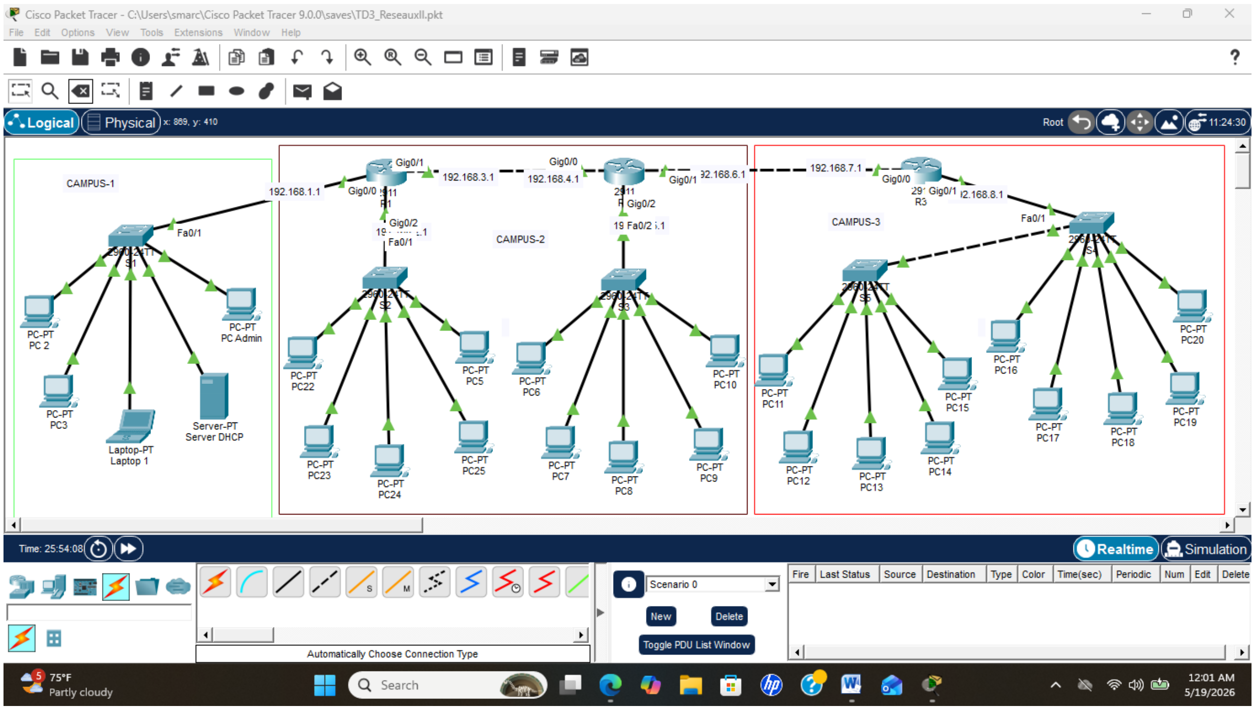
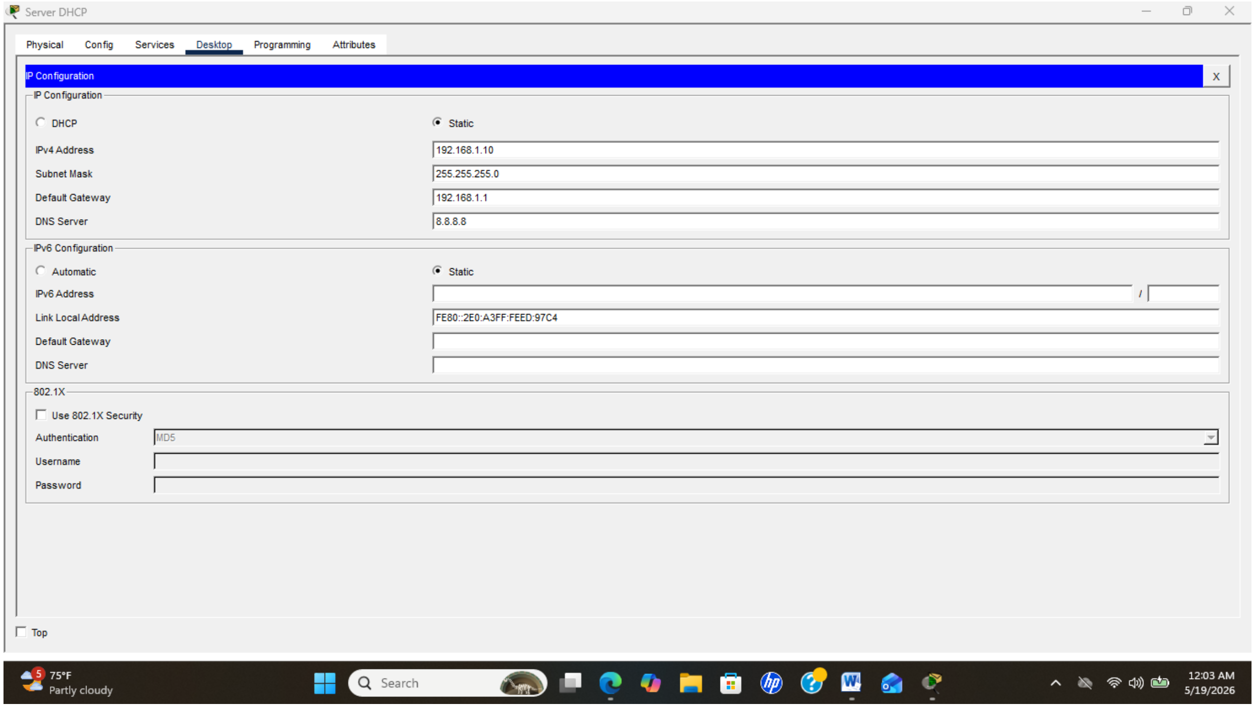
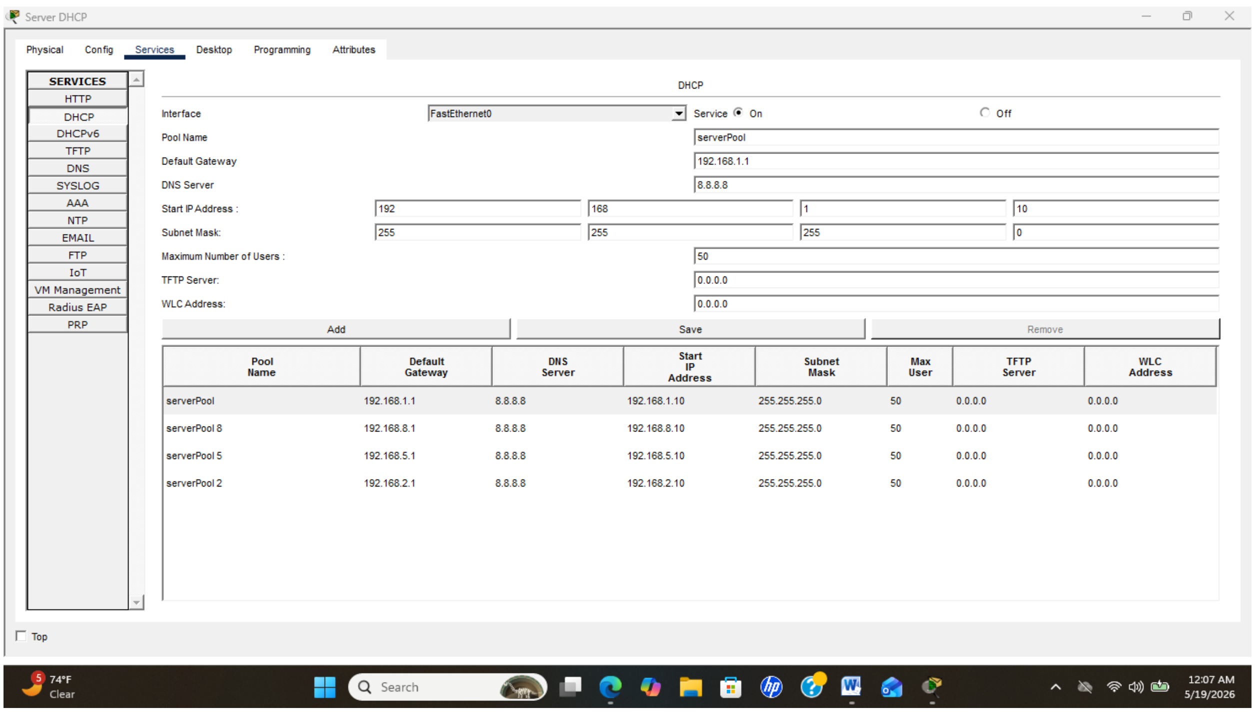


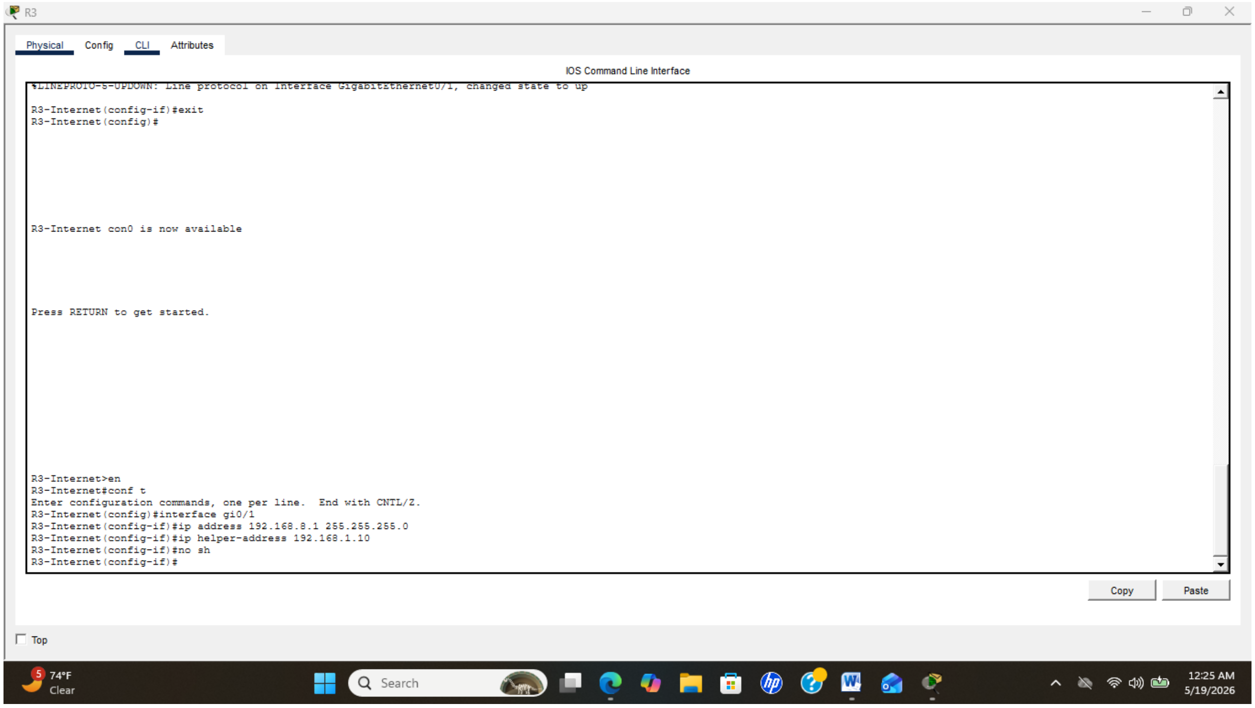
Figure 15 : Cette image atteste que la topologie du réseau a été correctement configurée.

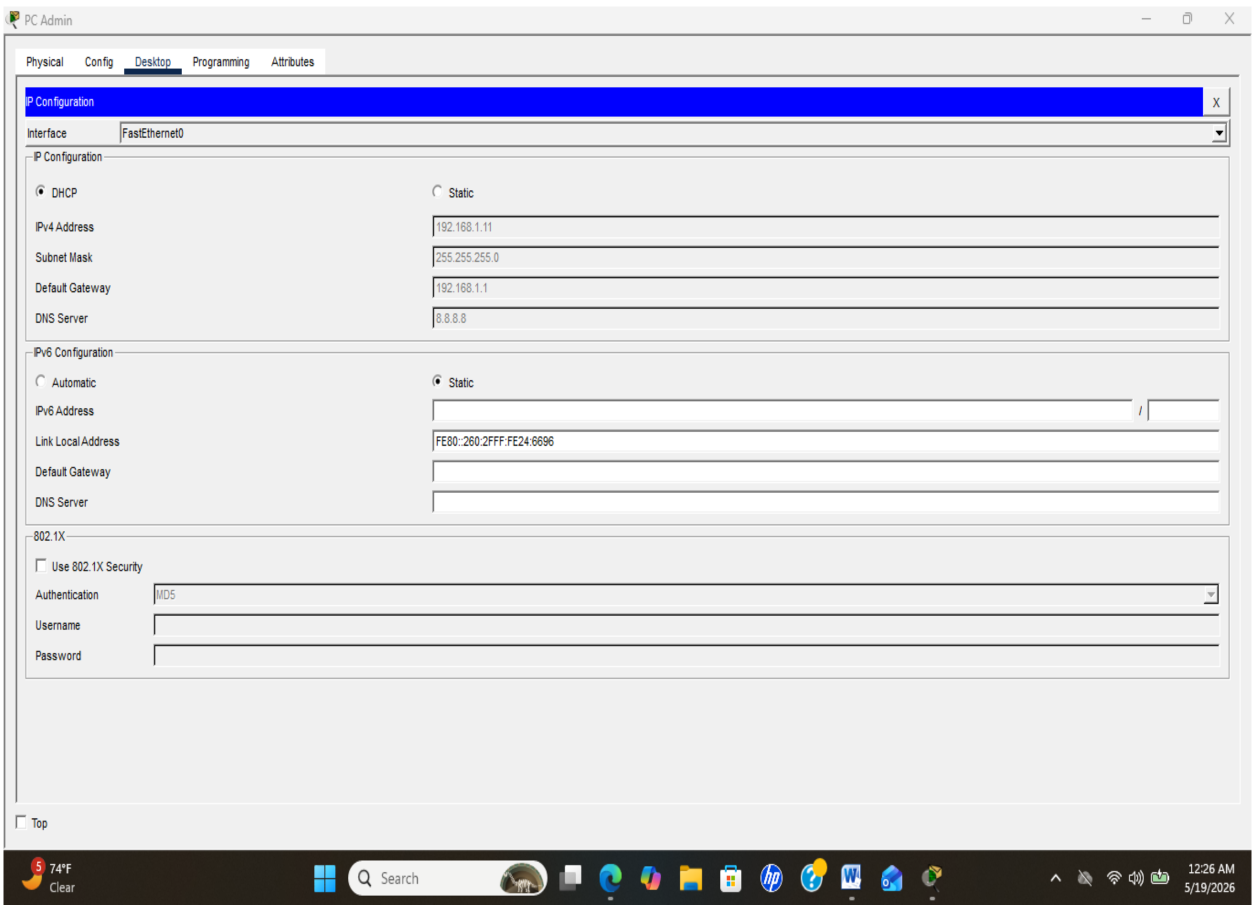
5 Configuration du Serveur DHCP





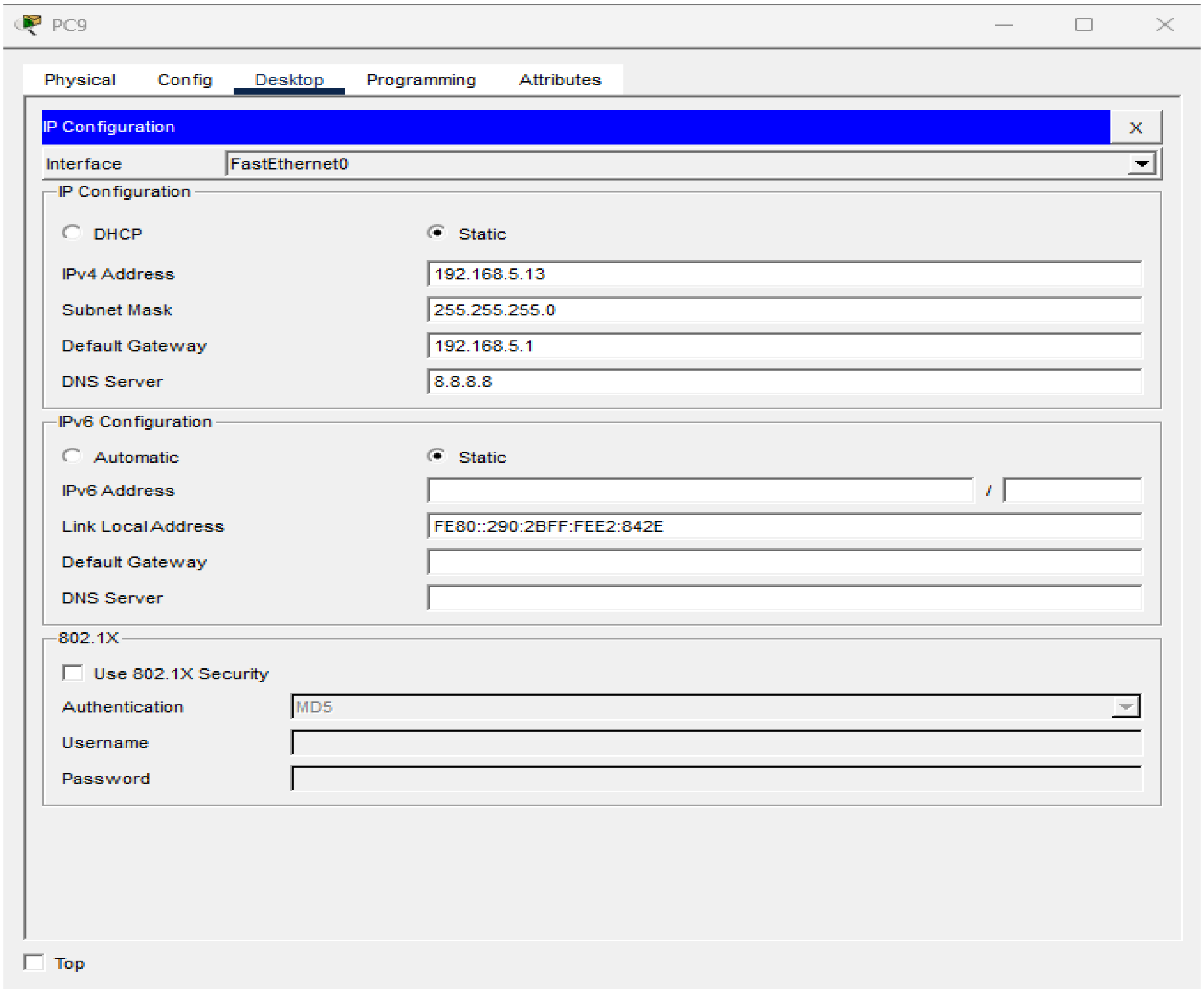
Figures 16 et 17 : Ces images montrent la configuration du serveur DHCP ainsi que l'attribution automatique des adresses IP aux équipements du réseau.

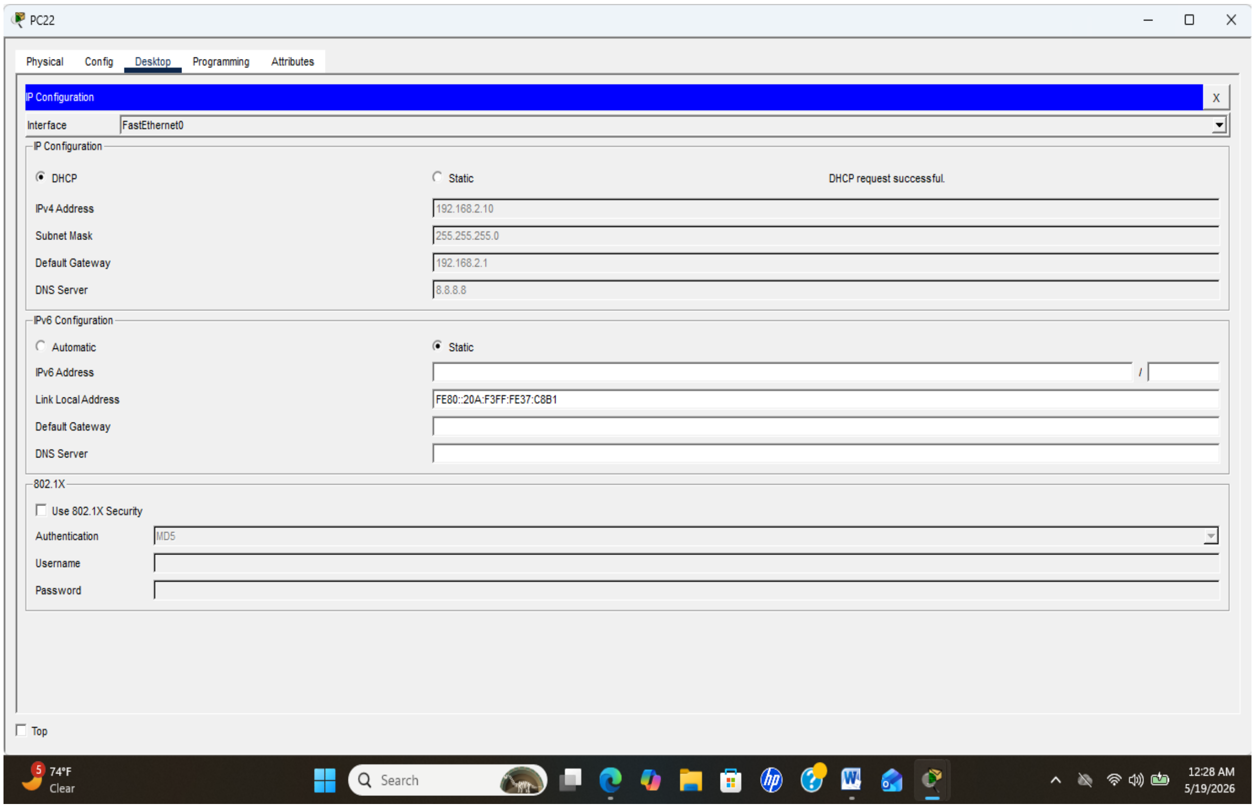

R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R3-Internet(config-if)#exit
R3-Internet(config)#
R3-Internet con0 is now available
Press RETURN to get started.
R3-Internet>en
R3-Internet#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3-Internet(config)#interface gi0/1
R3-Internet(config-if)#ip address 192.168.8.1 255.255.255.0
R3-Internet(config-if)#ip helper-address 192.168.1.10
R3-Internet(config-if)#no sh
R3-Internet(config-if)#
Copy
Paste
Top
74°F
Clear
Search
12:25 AM
5/19/2026


PC Admin
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FE24:6696
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
74°F
Clear
Search
12:26 AM
5/19/2026

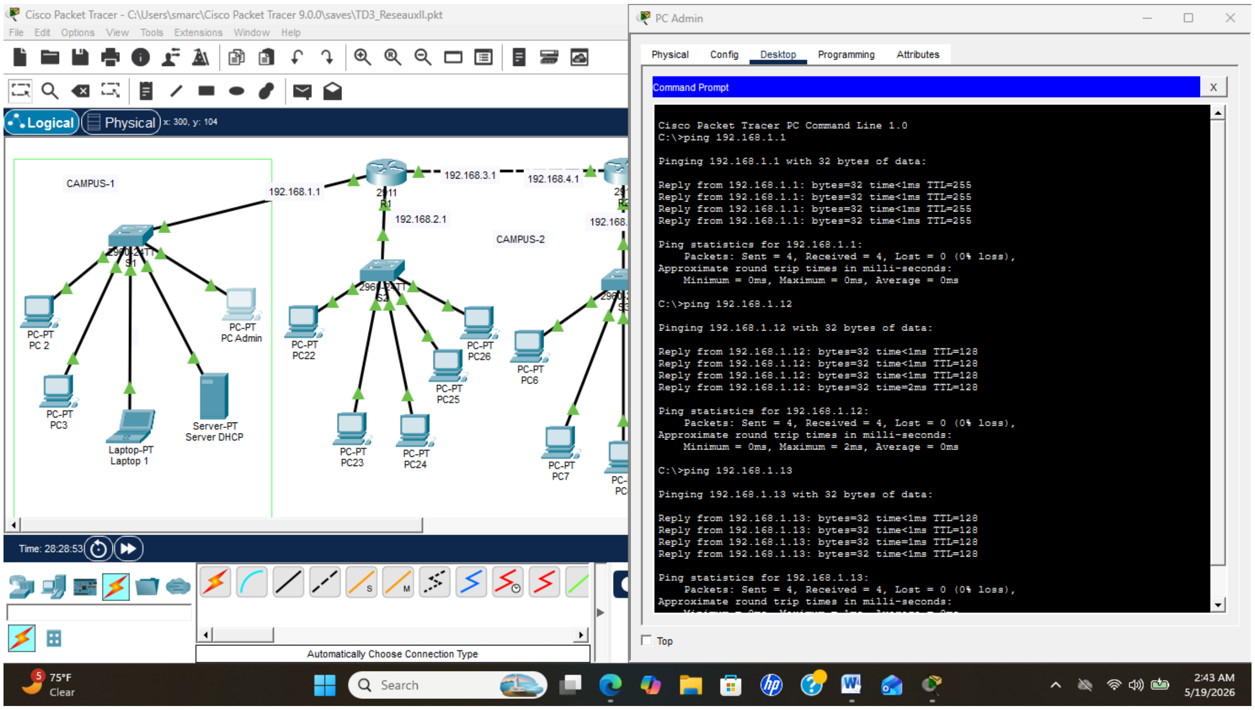
Remarque : Remarque : Après la configuration du serveur DHCP, la topologie a rencontré des difficultés de fonctionnement malgré plusieurs tentatives de correction (changement de câbles et vérification des connexions). Le serveur DHCP n'a pas réussi à attribuer correctement les adresses IP à tous les équipements. Ainsi, l'attribution des adresses IP s'est faite en deux parties : une partie automatiquement via le serveur DHCP et une autre configuration manuelle.

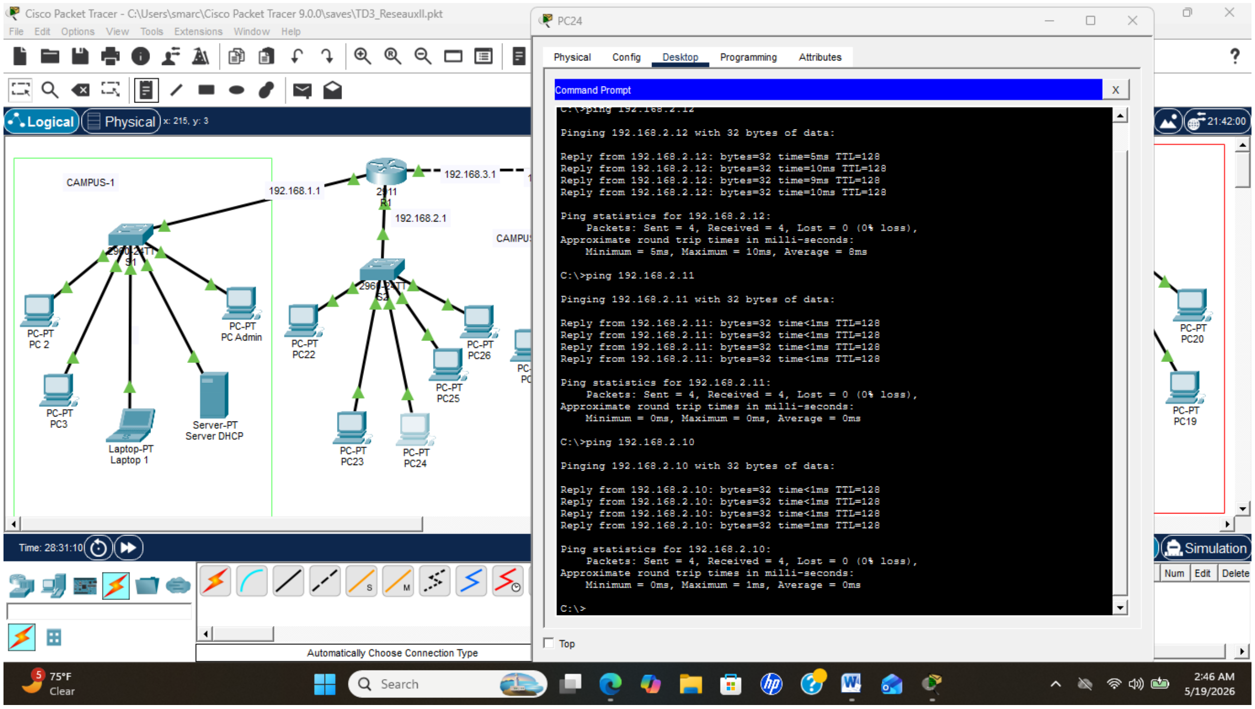
La configuration manuelle a concerné la plage d'adresses 192.168.5.1 à 192.168.8.1, où les équipements ont été configurés manuellement afin d'assurer la continuité du réseau.


PC9
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.5.13
Subnet Mask
255.255.255.0
Default Gateway
192.168.5.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:2BFF:FEE2:842E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

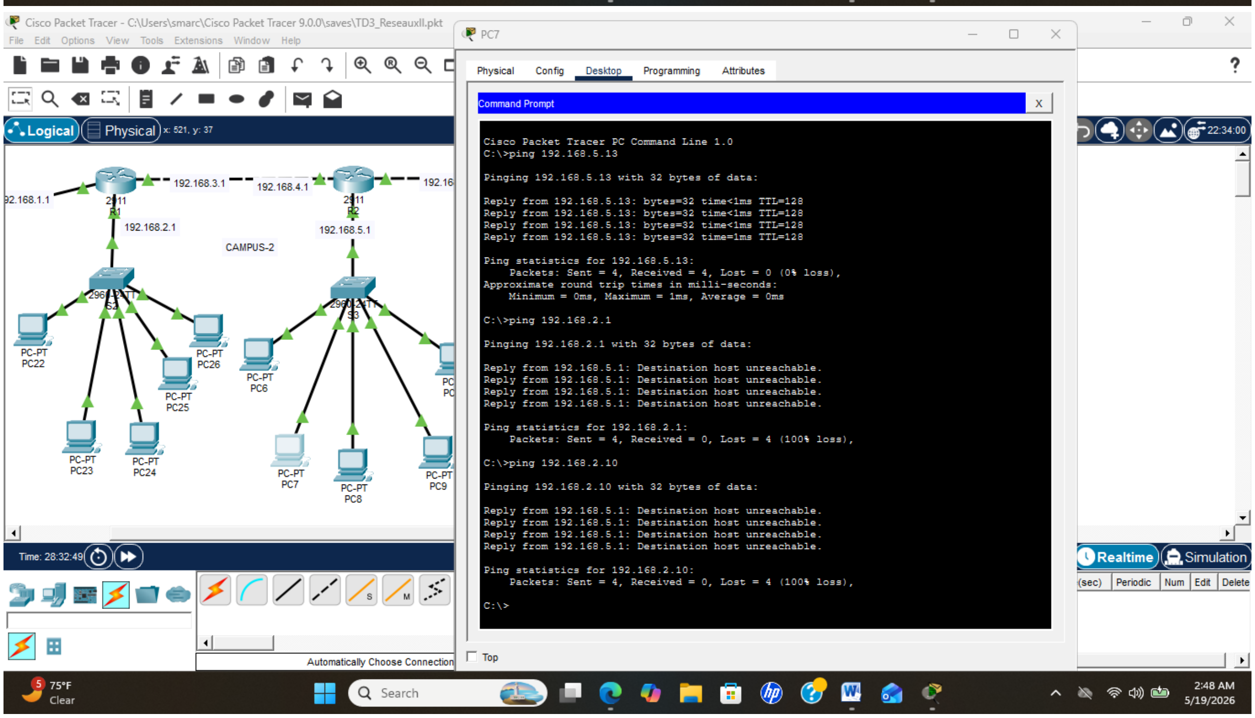


5.1 Test de connectivité

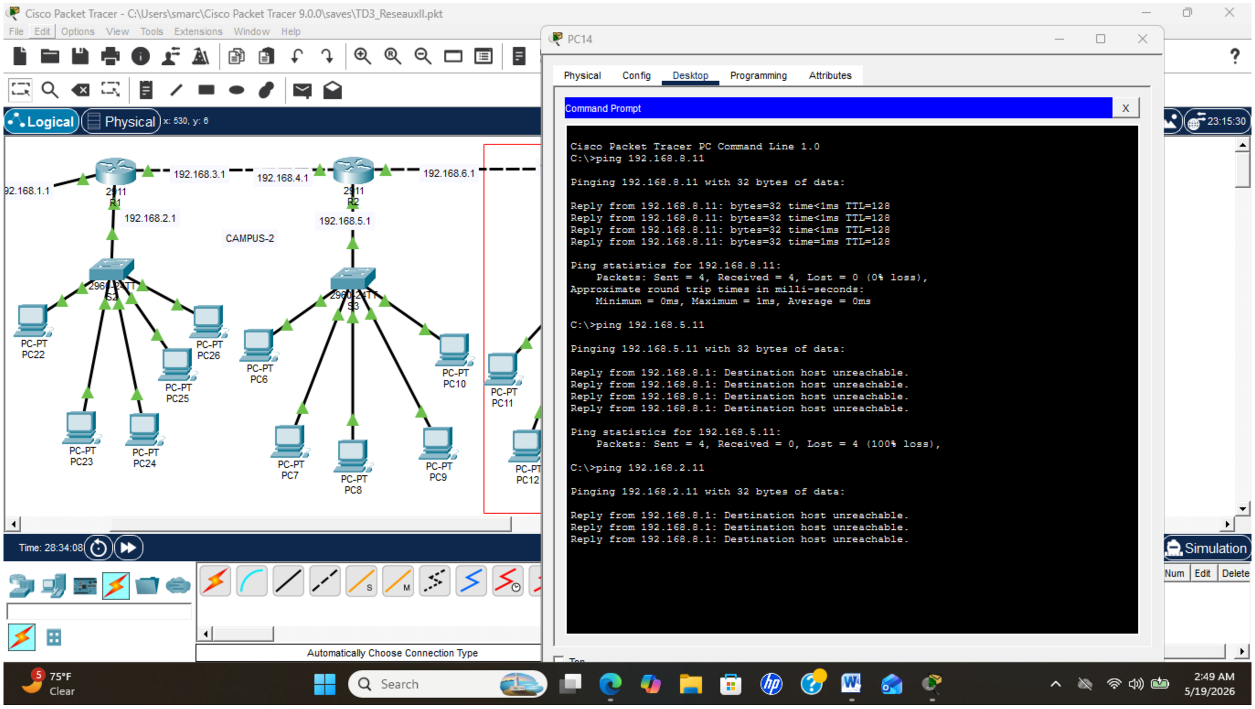




Cisco Packet Tracer - C:\Users\smarc\Cisco Packet Tracer 9.0.0\saves\TD3_ReseauxII.pkt
PC24
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Pinging 192.168.2.12 with 32 bytes of data:
Reply from 192.168.2.12: bytes=32 time=5ms TTL=128
Reply from 192.168.2.12: bytes=32 time=10ms TTL=128
Reply from 192.168.2.12: bytes=32 time=9ms TTL=128
Reply from 192.168.2.12: bytes=32 time=10ms TTL=128
Ping statistics for 192.168.2.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 10ms, Average = 8ms
C:\>ping 192.168.2.11
Pinging 192.168.2.11 with 32 bytes of data:
Reply from 192.168.2.11: bytes=32 time<1ms TTL=128
Reply from 192.168.2.11: bytes=32 time<1ms TTL=128
Reply from 192.168.2.11: bytes=32 time<1ms TTL=128
Reply from 192.168.2.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
CAMPUS-1
192.168.1.1
192.168.2.1
192.168.3.1
Server-PT
Server DHCP
Laptop-PT
Laptop 1
PC Admin



Cisco Packet Tracer - C:\Users\smarc\Cisco Packet Tracer 9.0.0\saves\TD3_ReseauxII.pkt
PC7
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.13
Pinging 192.168.5.13 with 32 bytes of data:
Reply from 192.168.5.13: bytes=32 time<1ms TTL=128
Reply from 192.168.5.13: bytes=32 time<1ms TTL=128
Reply from 192.168.5.13: bytes=32 time<1ms TTL=128
Reply from 192.168.5.13: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.5.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top
CAMPUS-2
192.168.3.1
192.168.4.1
192.168.2.1
192.168.5.1



Figures 22 à 25 : Ces images attestent que le test de connexion a été réussi avec succès (voir page 18 à 20).

5.2 Sécurité avancée

ACL étendues :

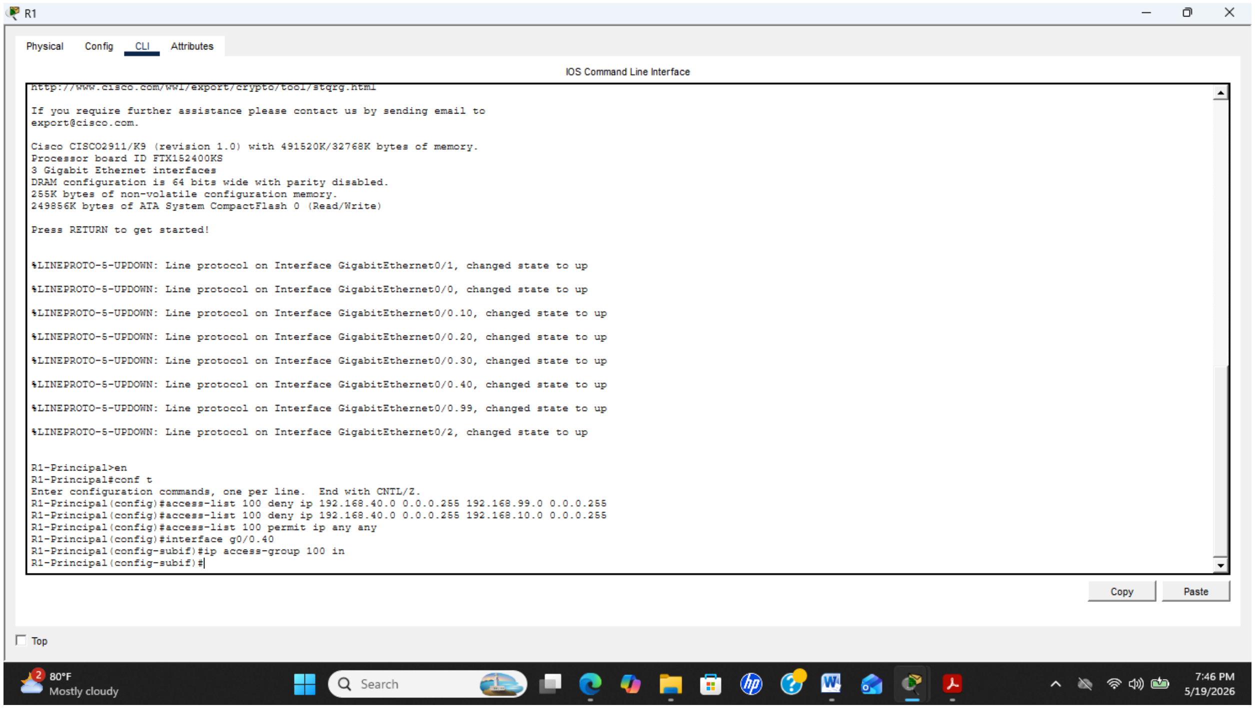
access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255

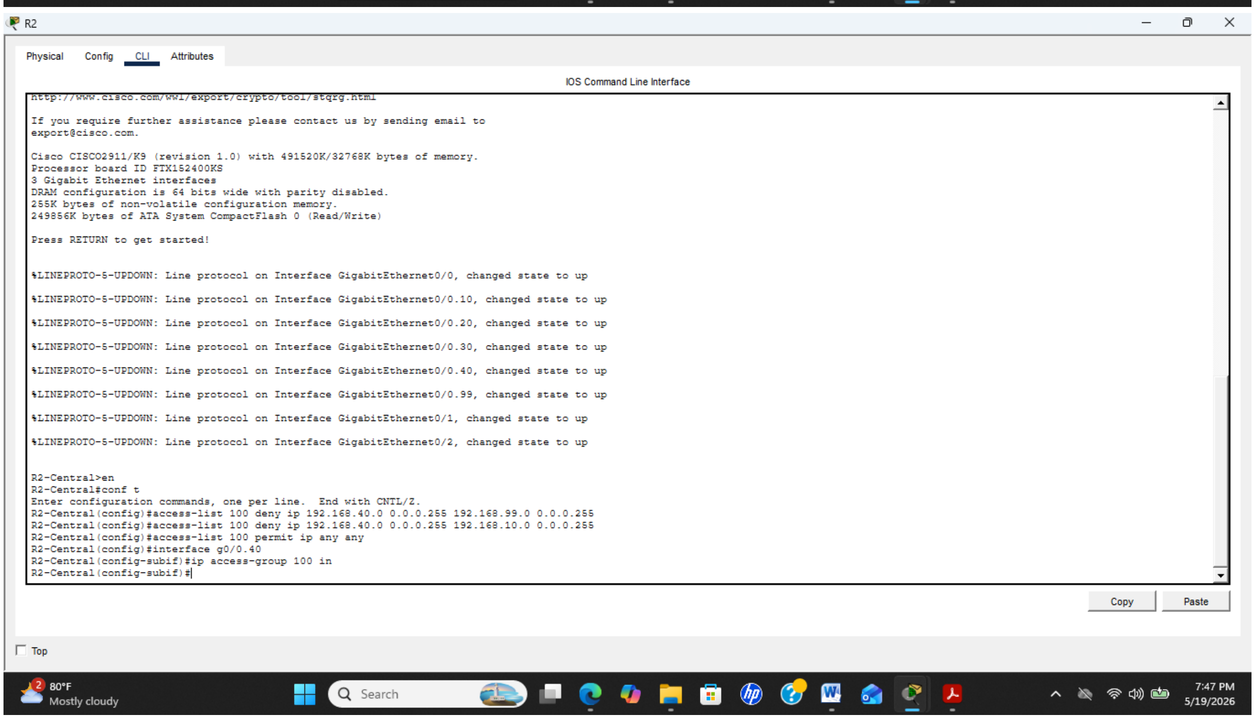
access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255

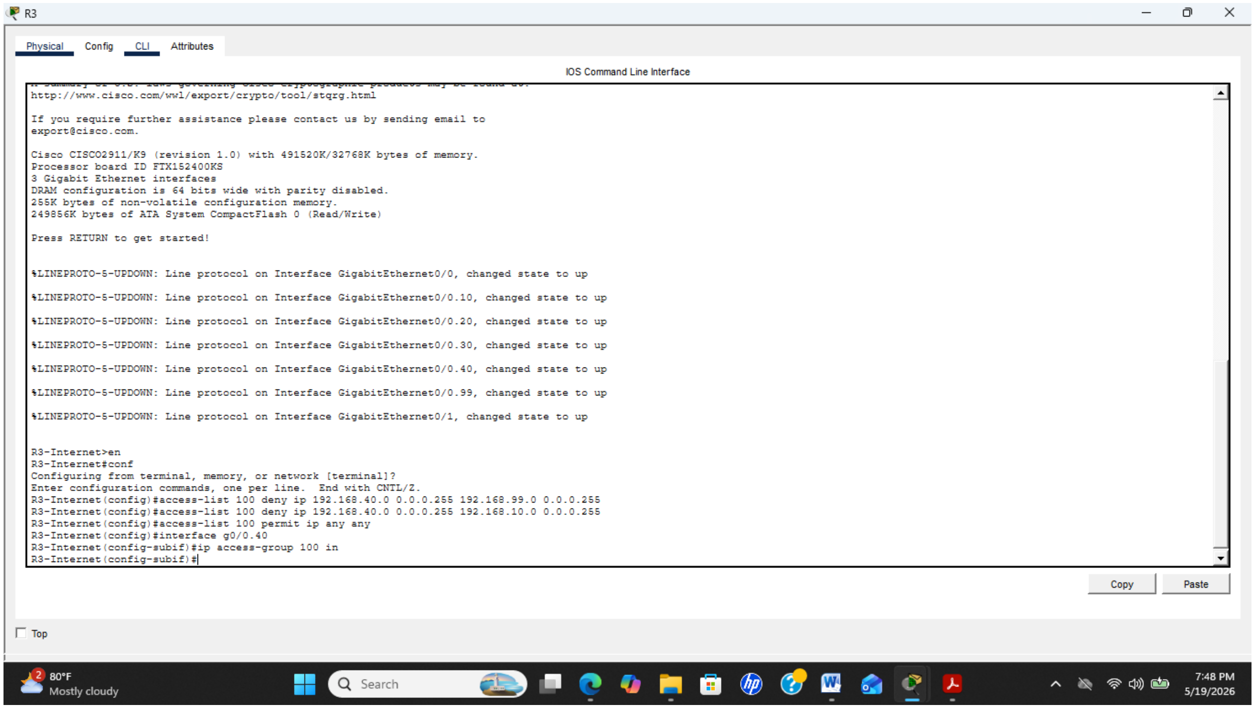
access-list 100 permit ip any any

interface g0/0.40

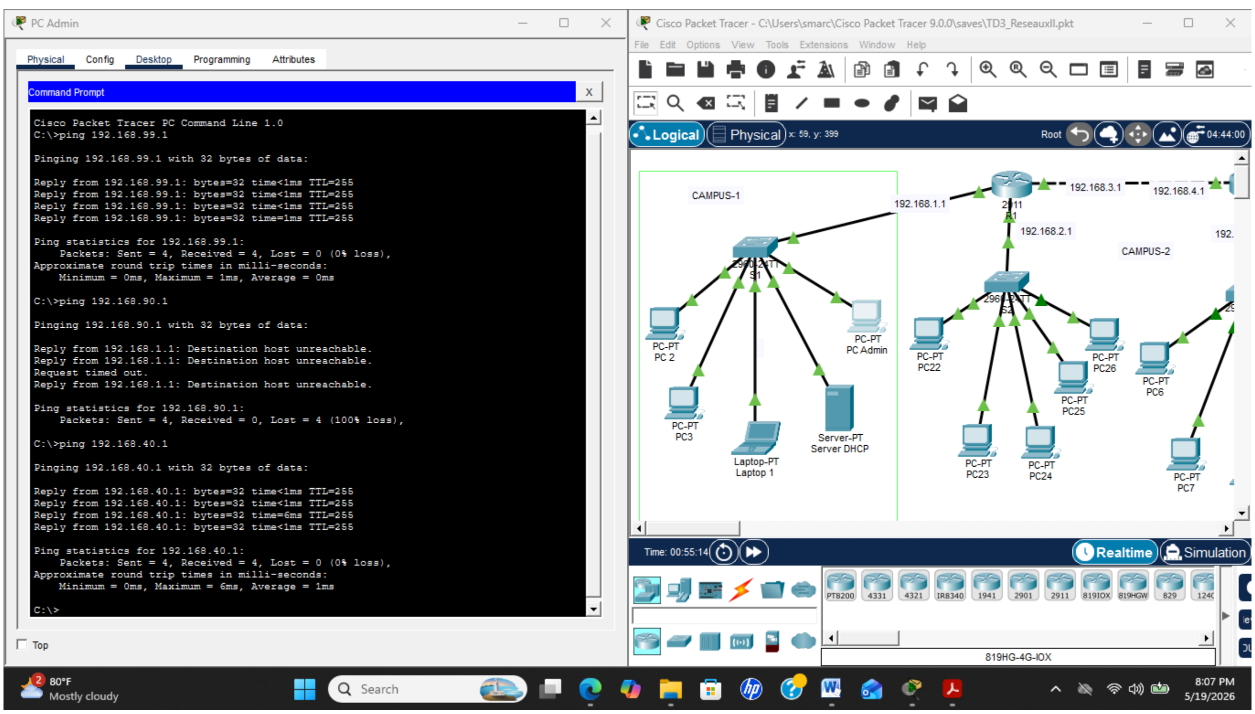
ip access-group 100 in


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
R1-Principal>en
R1-Principal#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1-Principal(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
R1-Principal(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
R1-Principal(config)#access-list 100 permit ip any any
R1-Principal(config)#interface g0/0.40
R1-Principal(config-subif)#ip access-group 100 in
R1-Principal(config-subif)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
7:46 PM
5/19/2026


R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
R2-Central>en
R2-Central#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2-Central(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
R2-Central(config)#access-list 100 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
R2-Central(config)#access-list 100 permit ip any any
R2-Central(config)#interface g0/0.40
R2-Central(config-subif)#ip access-group 100 in
R2-Central(config-subif)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
7:47 PM
5/19/2026



5.3 Test Sur PC Admin



Figures 26 à 29 : Ces captures d’écran montrent que le test avancé a été réussi avec succès (voir pages 21 à 22).

Port security sur les switches

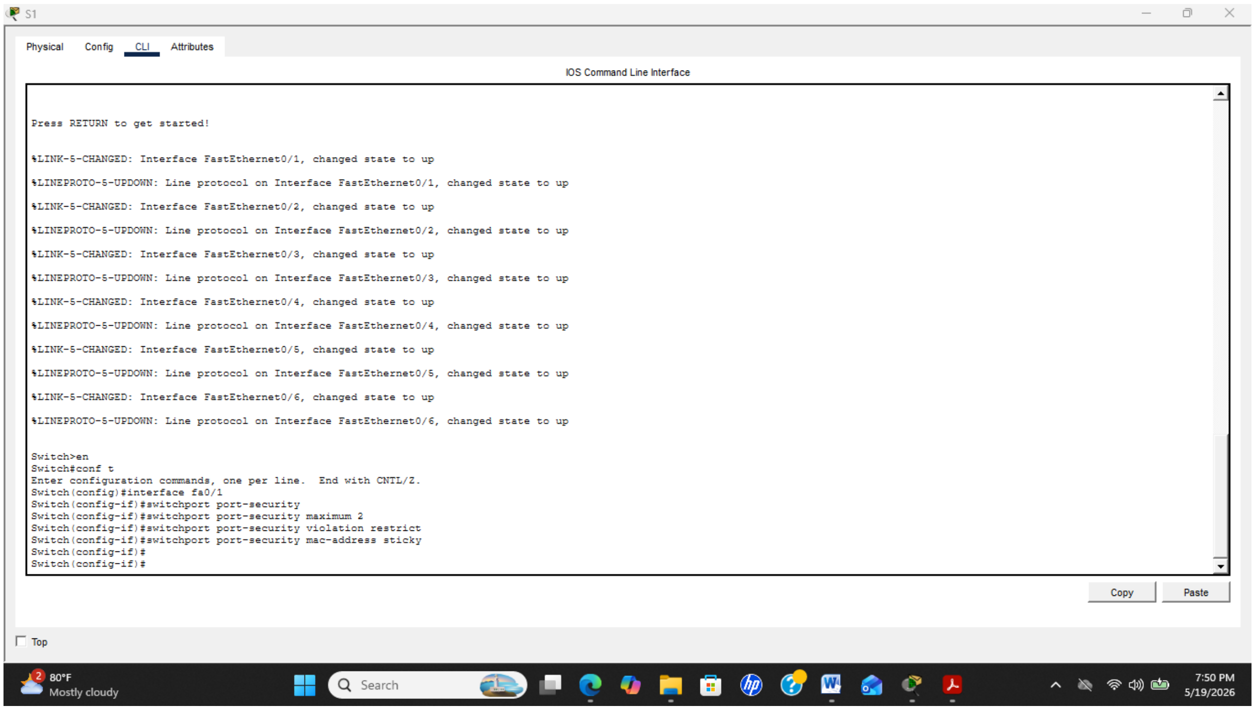
interface fa0/1

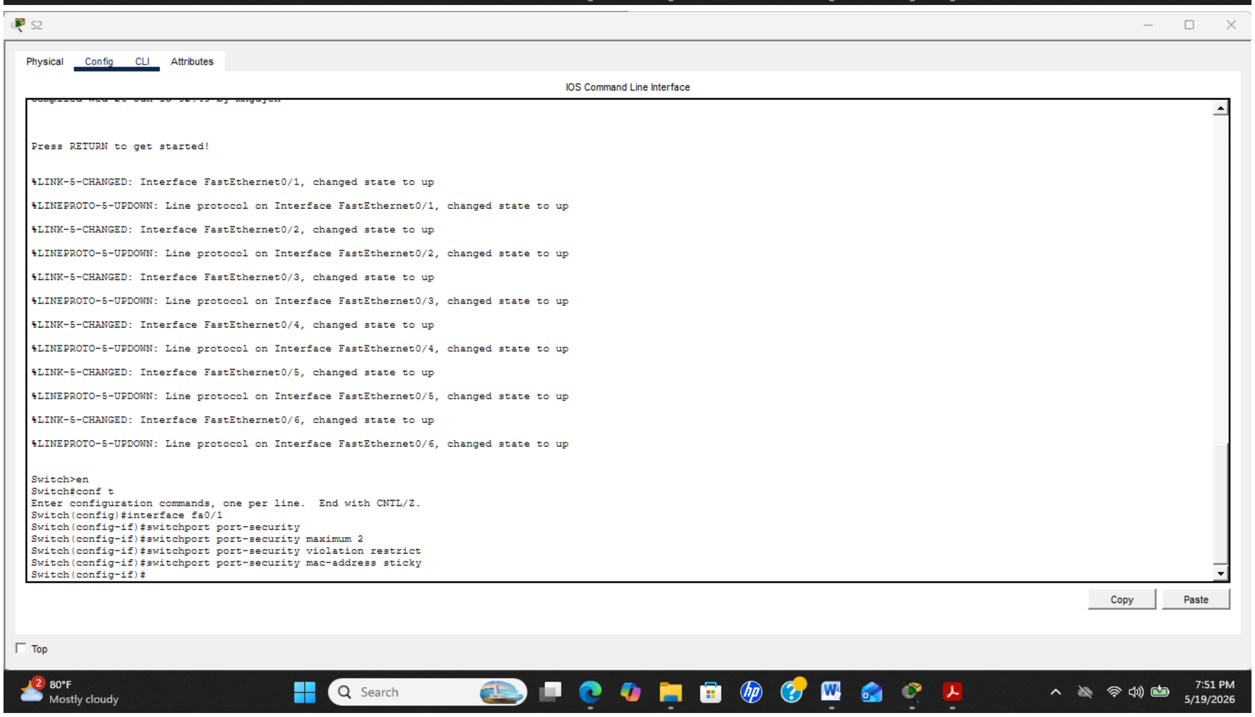
switchport port-security

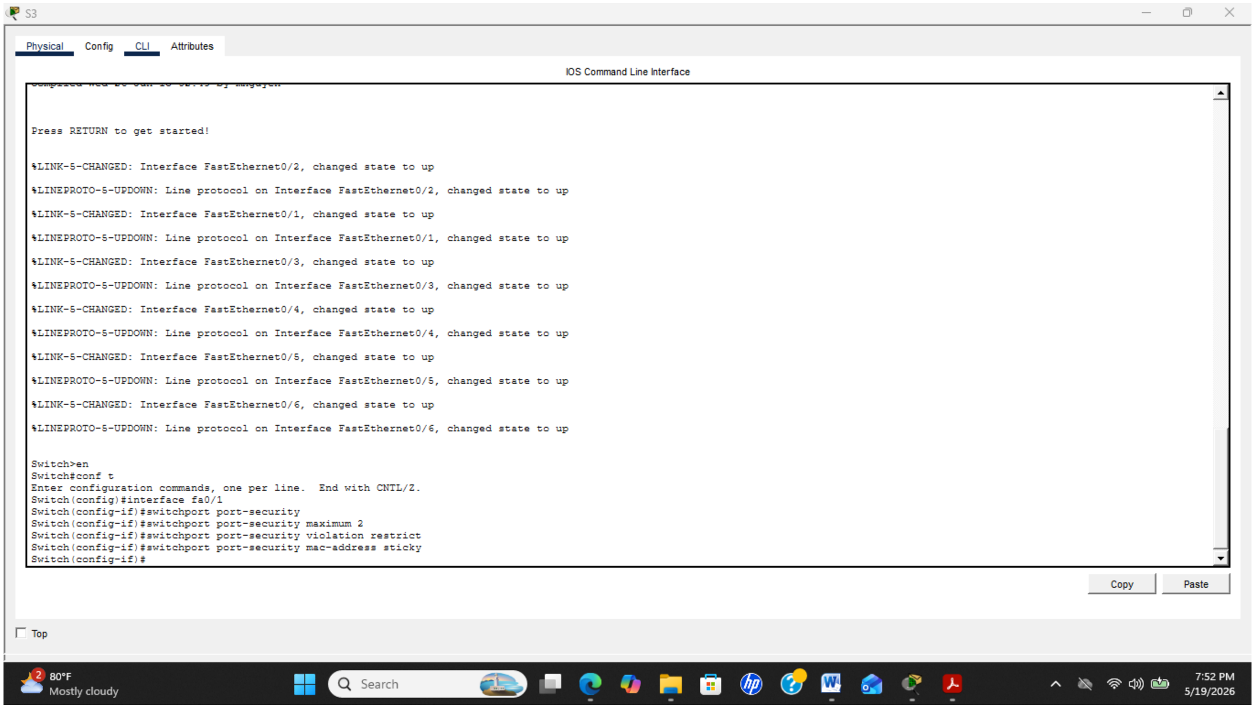
switchport port-security maximum 2

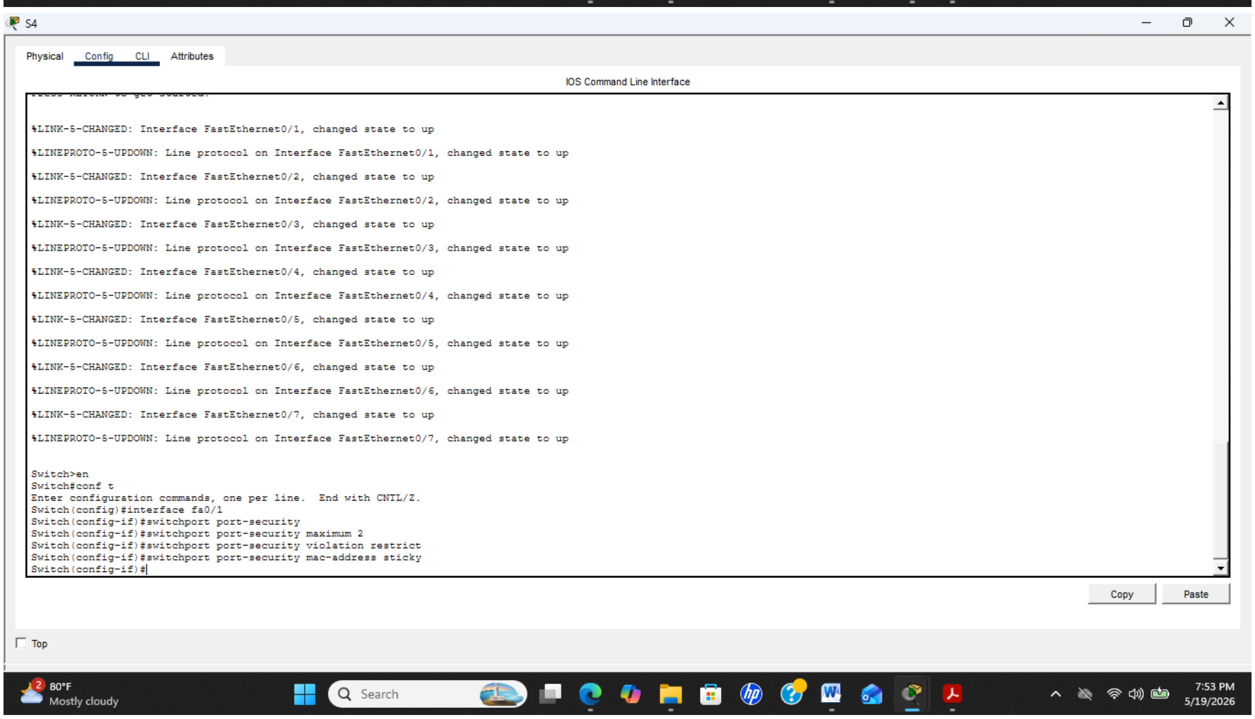
switchport port-security violation restrict

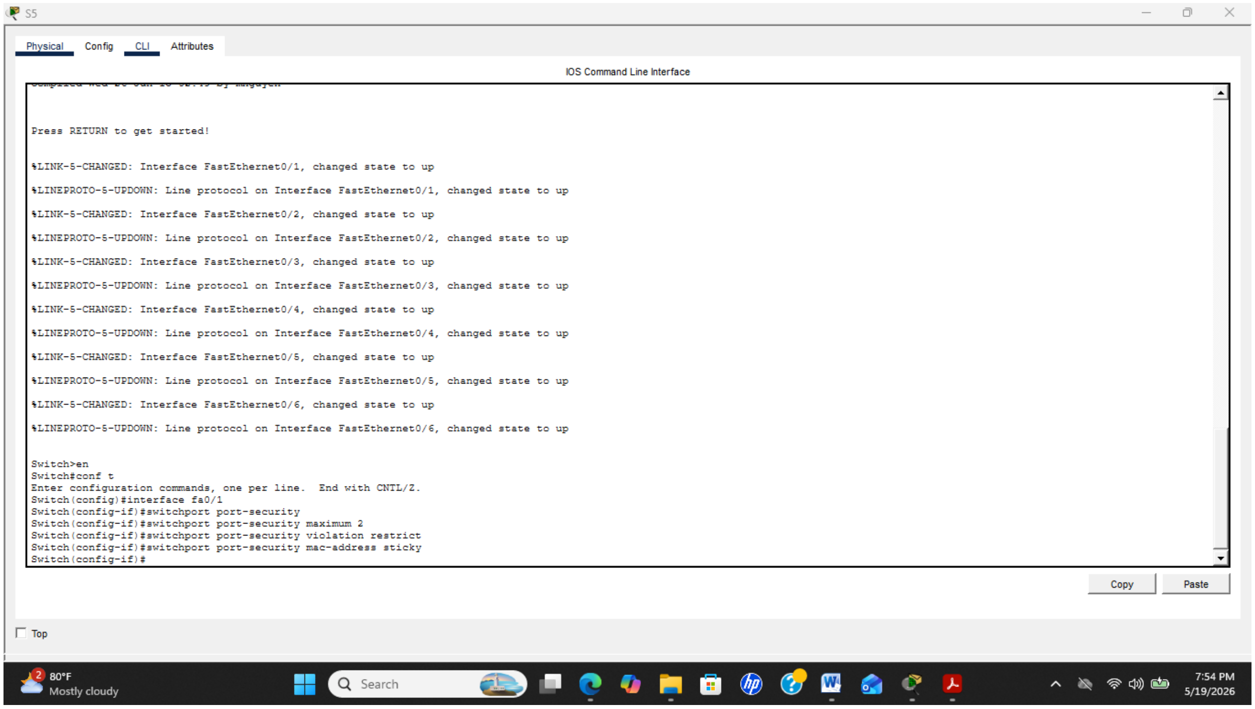
switchport port-security mac-address sticky






S3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 2
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
7:52 PM
5/19/2026


S4
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 2
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
7:53 PM
5/19/2026



Figures 30 à 34 : Ces images illustrent la configuration et le fonctionnement de la sécurité des ports (Port Security) sur les switches (voir pages 23 à 25).

SSH sur les routeurs

hostname R1

ip domain-name reseau.local

crypto key generate rsa

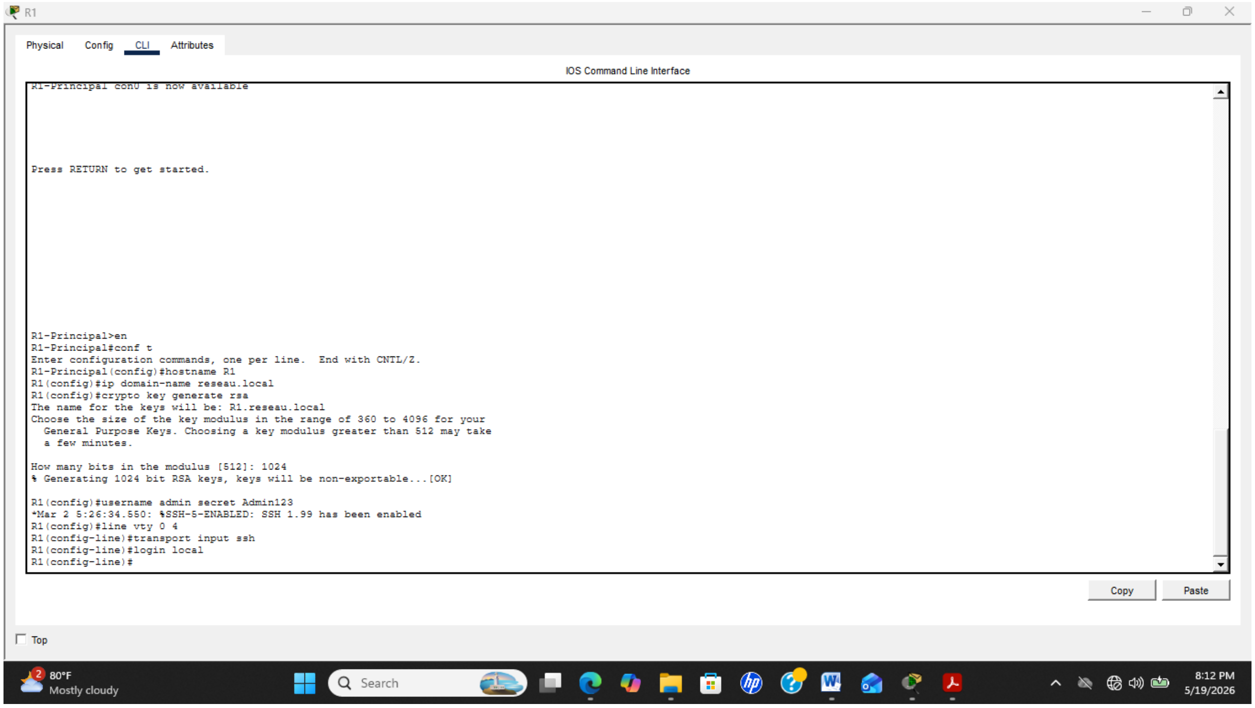
1024

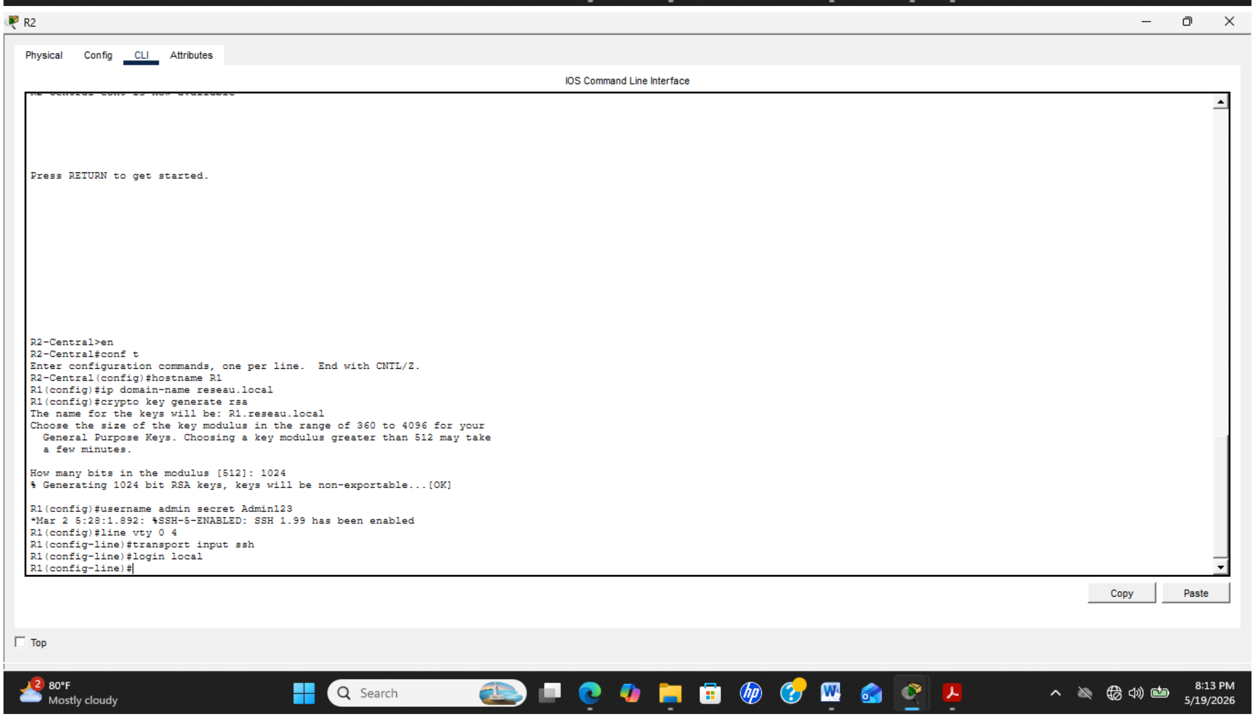
username admin secret Admin123

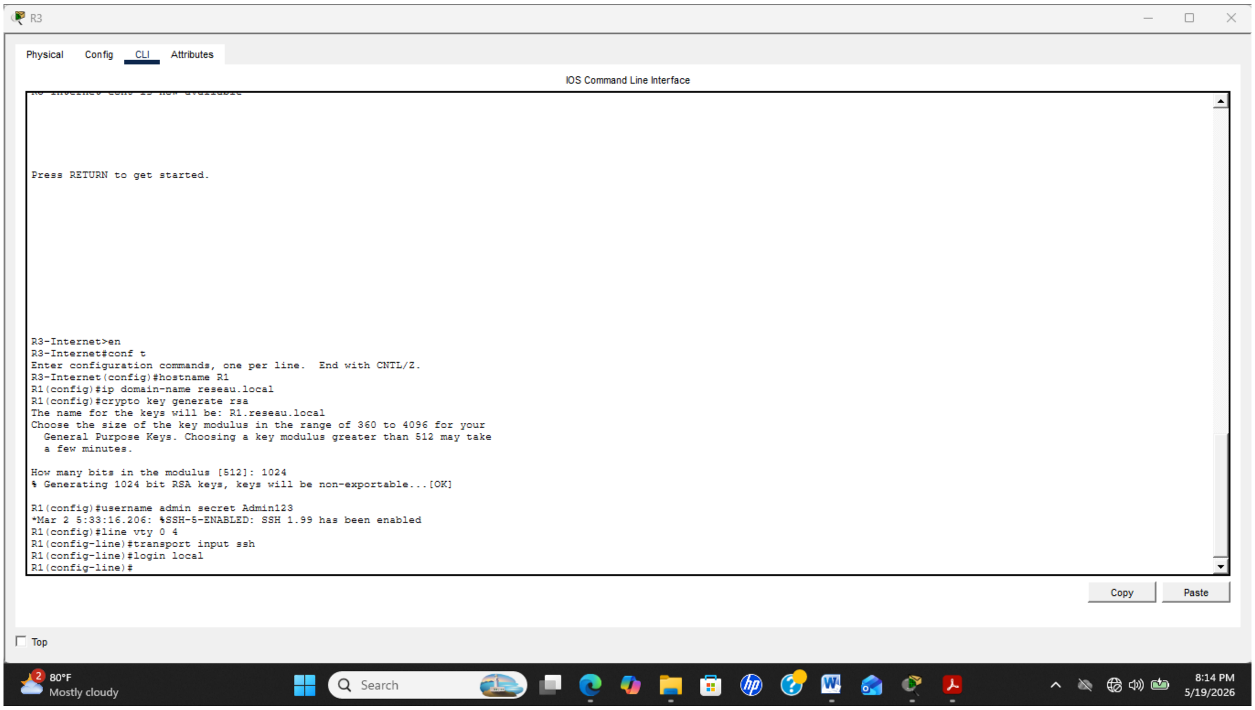
line vty 0 4

transport input ssh

login local


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
R1-Principal con0 is now available
Press RETURN to get started.
R1-Principal>en
R1-Principal#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1-Principal(config)#hostname R1
R1(config)#ip domain-name reseau.local
R1(config)#crypto key generate rsa
The name for the keys will be: R1.reseau.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
R1(config)#username admin secret Admin123
*Mar 2 5:26:34.550: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
8:12 PM
5/19/2026


R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
R2-Central>en
R2-Central#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2-Central(config)#hostname R1
R1(config)#ip domain-name reseau.local
R1(config)#crypto key generate rsa
The name for the keys will be: R1.reseau.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
R1(config)#username admin secret Admin123
*Mar 2 5:28:1.892: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#
Copy
Paste
Top
80°F
Mostly cloudy
Search
8:13 PM
5/19/2026



Figures 35 à 37 : Ces images attestent la configuration et le fonctionnement de SSH sur les routeurs (voir pages 26 à 27).

Scénarios de test

1. Ping Admin → Enseignants: OK
2. Ping Guest → Serveur : Bloqué
3. SSH : OK
4. Telnet : Désactivé
5. Port Security : OK

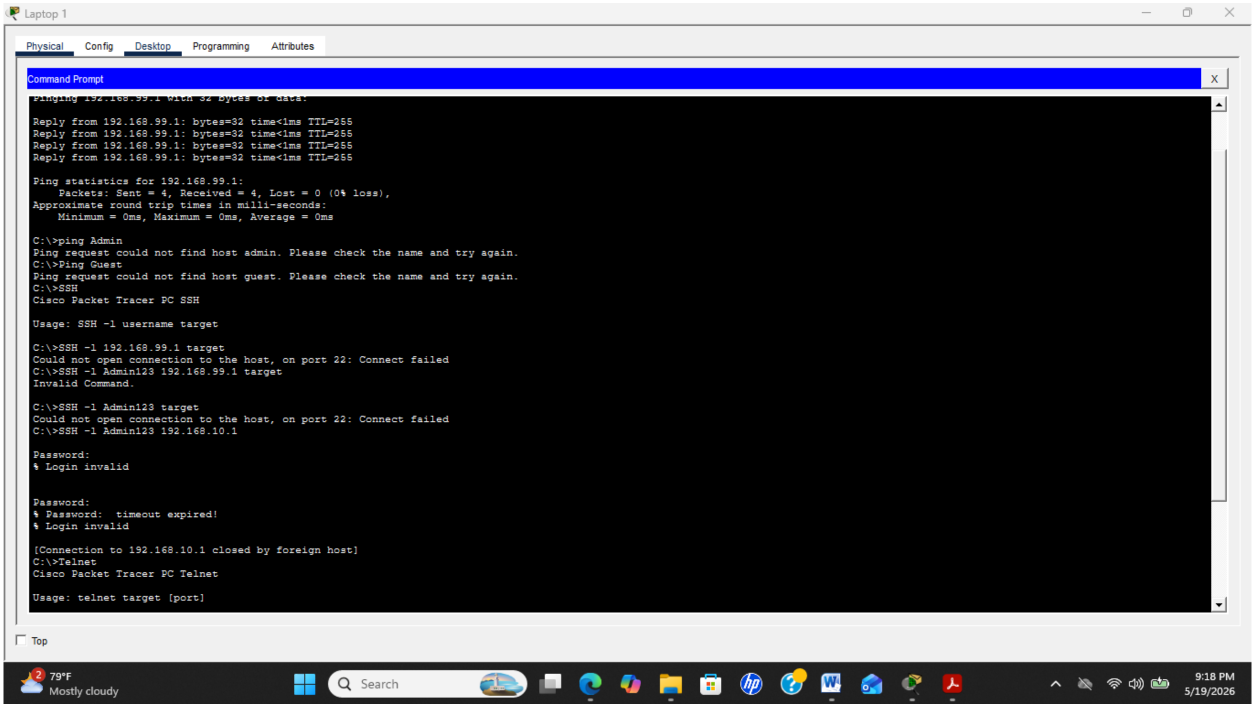


Figure 38 : Sur cette image, certains tests ont été réussis.

Conclusion

Ce projet a permis de mieux comprendre comment organiser et sécuriser un réseau de campus(IUS) à l'aide des VLANs, des ACL et de SSH. Ces éléments facilitent la gestion du réseau et renforcent la sécurité en contrôlant les accès entre les différents services.

Cependant, j'ai rencontré plusieurs difficultés, surtout au niveau de l'adressage IP des PCs et de la configuration du serveur DHCP. Il a été parfois compliqué de bien attribuer les bonnes adresses aux différents VLANs et de s'assurer que les machines reçoivent automatiquement une IP correcte via le DHCP.

Malgré ces difficultés, les tests finaux montrent que le réseau fonctionne correctement et que les règles de sécurité sont bien appliquées.